

Additive Universal Trust-Region Methods for Unconstrained and NC–SC Minimax Optimization

Abstract

We give three analyses of additive universal trust-region (UTR) methods. First, for unconstrained nonconvex optimization, we develop a certified-inexact adaptive clipped UTR whose outer scale update does not use a Hessian-Lipschitz constant. The inexact theorem is stated under explicit certificate conditions, whereas its exact-oracle specialization is fully parameter free and has trial scale $\bar{\sigma} = \max\{\sigma_0, \gamma_+ \sqrt{M/6}\}$; when $\sigma_0 = O(\sqrt{M})$, the total number of trust-region subproblems is

$$O\left(\Delta_F \sqrt{M} \epsilon^{-3/2} + \log_+ \frac{\|\nabla F(x_0)\|}{\epsilon} + \log_+ \frac{\sqrt{M}}{\sigma_0} + 1\right).$$

Thus all nonconstant search overhead is additive, and the leading term has the optimal dependence on the initial gap, Hessian smoothness, and tolerance. A fixed-scale inexact UTR estimate is developed only as a technical outer interface and supplies, in addition, the cubic-path budget needed below.

Second, for nonconvex–strongly-concave minimax optimization, we separate the outer method from a residual-based inner maximization module. Any module returning the prescribed inner residual yields

$$O\left(\Delta_P \kappa^{3/2} \sqrt{\rho} \epsilon^{-3/2} + \log_+ \frac{\|\nabla P(x_0)\|}{\epsilon} + 1\right)$$

outer iterations and corrected second-order oracle constructions. If the module has an accelerated $\sqrt{\kappa}$ residual-reduction bound, the leading inner-gradient complexity is

$$O\left((1 + \log \kappa) \Delta_P \kappa^2 \sqrt{\rho} \epsilon^{-3/2}\right).$$

A Newton-corrected value-function gradient makes the inner error quadratic; relative to the explicit raw-gradient analysis of Luo et al., this removes the leading multiplicative $\log(1/\epsilon)$.

Third, we instantiate the same residual interface with the parameter-free SCAR solver and give three fully observable outer constructions. An over-accurate adaptive SCAR–UTR retains the sharp outer rate but restores a leading inner $\log(1/\epsilon)$. An adaptive fixed-radius variant with persistent SCAR instead has leading outer and inner complexities

$$O(\Delta_P \kappa^{3/2} \sqrt{\rho} \epsilon^{-3/2}), \quad O(\Delta_P \kappa^2 \sqrt{\rho} \epsilon^{-3/2}),$$

respectively, with only additive instance-dependent lower-order terms. A gradient-scaled variant retains these leading bounds by amortizing its variable tracking work through a cubic-path budget. These parameter-free guarantees are asymptotic but class-uniform: after fixing common regularity constants (ℓ, μ, ρ) , one unknown threshold works for every payoff in the class and every initialization. For the over-accurate construction, the displayed sufficient threshold is $\Theta(\mu)$ as $\mu \downarrow 0$ with ℓ and ρ fixed. The algorithms neither use these constants nor certify that the requested tolerance lies below the relevant threshold.

1 Introduction

We first consider smooth unconstrained optimization $\min_x F(x)$. Approximate first-order stationarity does not exclude strict saddle points, so the target also controls the smallest Hessian eigenvalue. Additive universal trust-region (UTR) models obtain the standard $\epsilon^{-3/2}$ second-order scale using a quadratic subproblem whose regularization and radius depend on the gradient [1].

Our first contribution is a certified-inexact adaptive UTR that replaces the unknown scale $\sqrt{M/6}$ in the outer update by a persistent trial value. A multiplier-aware acceptance test controls both a certified objective proxy and gradient growth, while the persistent update amortizes rejected trials. Its exact-oracle specialization has sharp leading dependence on the initial objective gap, Hessian-Lipschitz constant, and target tolerance, with only additive search overhead. This is the consistently adaptive behavior emphasized in Hamad and Hinder [2] and, for cubic regularization, Fang et al. [3]. We also prove a fixed-scale inexact UTR estimate used only as an outer estimate: uniform gradient error can make a small interior step increase the objective, and the proof charges that increase to boundary decrease.

After completing both unconstrained analyses, we consider the nonconvex-strongly-concave value function $P(x) = \max_y f(x, y)$. Luo, Li, and Chen apply an inexact cubic-regularized Newton method to this problem [4]. Write $\kappa = \ell/\mu$, let ρ be the Hessian-Lipschitz constant of f , and set $\Delta_P = P(x_0) - \inf_x P(x)$. Their outer complexity is

$$O\left(\Delta_P \kappa^{3/2} \sqrt{\rho} \epsilon^{-3/2}\right),$$

and their stated gradient complexity is $\tilde{O}(\Delta_P \kappa^2 \sqrt{\rho} \epsilon^{-3/2})$. In the small- ϵ branch of their explicit worst-case bound, the raw estimator $\nabla_x f(x, y)$ enforces an $O(\epsilon/\ell)$ inner distance and produces a leading $\log(1/\epsilon)$.

The minimax specialization here instead uses a Newton-corrected value-function gradient. Its inner error is quadratic, so an accelerated inner module only needs a distance of order $\sqrt{\epsilon}$. Our three contributions can be summarized as follows.

- (i) For unconstrained optimization, the certified-inexact adaptive UTR admits value, gradient, and Hessian certificates and can therefore be instantiated for the minimax value function whenever such certificates are supplied. As an exact-oracle by-product, the method does not use M and, whenever $\sigma_0 = O(\sqrt{M})$, solves at most

$$O\left(\Delta_F \sqrt{M} \epsilon^{-3/2} + \log_+ \frac{\|\nabla F(x_0)\|}{\epsilon} + \log_+ \frac{\sqrt{M}}{\sigma_0} + 1\right)$$

trial subproblems. Its returned point is an $O(\epsilon, \sqrt{M}\epsilon)$ -SOSP, and no logarithm multiplies the leading term, in the consistently adaptive sense emphasized by Hamad and Hinder [2], Fang et al. [3].

- (ii) Gradient-norm regularized trust-region models have already been considered for NC-SC minimax problems by Wang and Xu [5], while Luo et al. [4] analyzes a cubic-regularized method. Our contribution is an inexact-oracle reduction based on a Newton-corrected envelope gradient and a modular inner residual interface. It gives

$$O\left(\Delta_P \kappa^{3/2} \sqrt{\rho} \epsilon^{-3/2} + \log_+ \frac{\|\nabla P(x_0)\|}{\epsilon} + 1\right)$$

corrected second-order oracle constructions and, for an accelerated inner module, leading inner-gradient complexity

$$O\left((1 + \log \kappa) \Delta_P \kappa^2 \sqrt{\rho} \epsilon^{-3/2}\right).$$

At the level of the proved worst-case bounds, this improves the raw plug-in-gradient analysis of Luo et al. [4] by one leading factor $\log(1/\epsilon)$.

- (iii) We combine the same residual interface with SCAR to remove all problem parameters from the executable minimax algorithms. The over-accurate adaptive SCAR-UTR has sharp leading second-order complexity but pays a leading inner $\log(1/\epsilon)$. The fixed-radius persistent-SCAR variant removes that logarithm and has leading outer and inner bounds

$$O(\Delta_P \kappa^{3/2} \sqrt{\rho} \epsilon^{-3/2}), \quad O(\Delta_P \kappa^2 \sqrt{\rho} \epsilon^{-3/2}).$$

A gradient-scaled persistent-SCAR variant has the same leading bounds, with variable tracking work controlled by a gradient-level potential and a cubic-path estimate. These parameter-free results are asymptotic but class-uniform: for fixed common regularity constants, one small-tolerance threshold works over the entire problem class and all initializations. For the over-accurate construction, the displayed sufficient threshold scales as $\Theta(\mu)$ for fixed ℓ and ρ . The thresholds are not observable, so the results are not a posteriori finite-accuracy certificates.

Organization. Section 2 introduces the unconstrained problem. Section 3 records the fixed-scale inexact estimates used later in the minimax proof, and Section 4 gives a certified-inexact adaptive method and its complete trial complexity. Section 5 then introduces the minimax setting, the residual-based inner module, and the corrected oracle reduction. Sections 6 to 8 give the parameter-free SCAR implementations. Section A collects the deferred proofs, and Section B explains the class-uniform quantifiers and the limitation of residual-only certificates.

2 Unconstrained problem and preliminaries

We first consider

$$\min_{x \in \mathbb{R}^n} F(x), \tag{2.1}$$

where $F^* := \inf_x F(x) > -\infty$ and $\Delta_F := F(x_0) - F^*$.

Assumption 2.1 (Hessian regularity). *The function F is twice continuously differentiable and, for some $M > 0$,*

$$\|\nabla^2 F(x) - \nabla^2 F(z)\| \leq M\|x - z\| \quad \forall x, z \in \mathbb{R}^n. \tag{2.2}$$

2.1 Notation and approximate stationarity

The norm $\|\cdot\|$ is Euclidean for vectors and spectral for matrices. For a symmetric matrix H , $\lambda_{\min}(H)$ denotes its smallest eigenvalue. We write $\log_+(t) = \max\{\log t, 0\}$ for $t > 0$ and set $\log_+(0) = 0$. We also write $\mathbb{N}_0 := \{0, 1, 2, \dots\}$. The notation $\tilde{O}(\cdot)$ suppresses logarithmic factors.

Definition 2.1 (Approximate stationary points). *For a twice differentiable function F and tolerances $\epsilon, \epsilon_g, \epsilon_H > 0$, a point x is an ϵ -first-order stationary point if $\|\nabla F(x)\| \leq \epsilon$. It is an (ϵ_g, ϵ_H) -second-order stationary point if*

$$\|\nabla F(x)\| \leq \epsilon_g, \quad \lambda_{\min}(\nabla^2 F(x)) \geq -\epsilon_H.$$

Our generic target is an $(\epsilon, \sqrt{M\epsilon})$ -second-order stationary point when $\nabla^2 F$ is M -Lipschitz.

We repeatedly use the following standard consequences of Hessian Lipschitz continuity.

Lemma 2.1 (Second-order Taylor bounds). *If $\nabla^2 F$ is M -Lipschitz, then, for all x, d ,*

$$\|\nabla F(x + d) - \nabla F(x) - \nabla^2 F(x)d\| \leq \frac{M}{2}\|d\|^2. \tag{2.3a}$$

$$F(x+d) \leq F(x) + \nabla F(x)^T d + \frac{1}{2} d^T \nabla^2 F(x) d + \frac{M}{6} \|d\|^3. \quad (2.3b)$$

These estimates are standard; see Nesterov [6, Lemma 1.2.4].

Lemma 2.2 (Global optimality for a quadratic trust-region problem). *Let $g \in \mathbb{R}^n$, let $B \in \mathbb{R}^{n \times n}$ be symmetric, and let $R > 0$. A vector d is a global solution of*

$$\min_{\|u\| \leq R} g^T u + \frac{1}{2} u^T B u$$

if and only if there is a multiplier $\lambda \geq 0$ such that

$$(B + \lambda I)d = -g, \quad B + \lambda I \succeq 0, \quad \|d\| \leq R, \quad \lambda(\|d\| - R) = 0. \quad (2.4)$$

In particular, $\lambda > 0$ implies $\|d\| = R$.

This characterization is classical; see, for example, Luenberger and Ye [7, Section 8.7]. All trust-region subproblems below are solved globally, so Lemma 2.2 applies throughout.

3 Fixed-scale inexact UTR estimates

This section isolates the fixed-scale estimates used later in the minimax reduction. These estimates analyze repeated, globally solved additive UTR subproblems; they are not presented as a separate algorithmic contribution. All technical proofs are deferred to Section A.2.

Assumption 3.1 (Certified uniform derivative accuracy). *Fix a target tolerance $\epsilon > 0$. At every iterate, the inexact oracle returns $g_k \in \mathbb{R}^n$, a symmetric $H_k \in \mathbb{R}^{n \times n}$, and an observable Hessian-error certificate $\xi_k \geq 0$ satisfying*

$$\|g_k - \nabla F(x_k)\| \leq \frac{\epsilon}{8192}, \quad \|H_k - \nabla^2 F(x_k)\| \leq \xi_k \leq \frac{1}{64} \sqrt{\frac{M}{6}} \sqrt{\epsilon}. \quad (3.1)$$

Set

$$h_k := \max\{\|g_k\|, \epsilon\}. \quad (3.2)$$

At iterate x_k , terminate if

$$\|g_k\| \leq \epsilon, \quad \lambda_{\min}(H_k) \geq -\sqrt{M}\epsilon + \xi_k. \quad (3.3)$$

Otherwise, compute a global solution d_k of

$$\begin{aligned} \min_{d \in \mathbb{R}^n} \quad & g_k^T d + \frac{1}{2} d^T \left(H_k + \xi_k I + 2\sqrt{\frac{M}{6}} \sqrt{h_k} I \right) d, \\ \text{s. t.} \quad & \|d\| \leq \frac{\sqrt{h_k}}{4\sqrt{M/6}}, \end{aligned} \quad (3.4)$$

and set $x_{k+1} = x_k + d_k$. This is a certificate-padded variant of the additive UTR subproblem in Jiang et al. [1], evaluated with the derivative estimates (g_k, H_k) and the certificate ξ_k ; the trust-region subproblem itself is solved exactly. The stopping rule is worst-case sharp given only the scalar certificate ξ_k : $\lambda_{\min}(H_k) \geq -\sqrt{M}\epsilon + \xi_k$ is the weakest condition on H_k that guarantees $\lambda_{\min}(\nabla^2 F(x_k)) \geq -\sqrt{M}\epsilon$ using only $\|H_k - \nabla^2 F(x_k)\| \leq \xi_k$. The matching shift $\xi_k I$ absorbs the same uncertainty in the model. The remaining clean coefficients 2 and 1/4 reduce the explicit worst-case iteration constant. Optimizing decimal coefficients would depend on the particular counting potential and would not define an intrinsically more aggressive method.

3.1 Complexity estimates

The one-step descent, clipped-gradient recurrence, and finite-prefix counting estimates are stated and proved in [Section A.2](#). Their two consequences needed in the minimax analysis are recorded next.

Theorem 3.1 (Fixed-scale inexact UTR estimate). *Fix x_0 and $\epsilon > 0$, and define $F^* := \inf_x F(x) > -\infty$, $\Delta_F := F(x_0) - F^*$, and $h_0 := \max\{\|g_0\|, \epsilon\}$. Suppose [Assumptions 2.1](#) and [3.1](#) hold. Starting from x_0 , repeatedly apply the stopping rule [\(3.3\)](#) and the globally solved subproblem [\(3.4\)](#). This procedure terminates after at most*

$$65\sqrt{\frac{M}{6}} \Delta_F \epsilon^{-3/2} + 3 \log_+ \frac{h_0}{\epsilon} + 2 \quad (3.5)$$

nonterminal iterations, equivalently trust-region subproblem solves. In particular, this is $O(\Delta_F \sqrt{M} \epsilon^{-3/2} + \log_+(h_0/\epsilon) + 1)$. Its output $\hat{x}$ satisfies

$$\|\nabla F(\hat{x})\| \leq \frac{8193}{8192} \epsilon, \quad \lambda_{\min}(\nabla^2 F(\hat{x})) \geq -\sqrt{M} \epsilon. \quad (3.6)$$

Replacing the internal tolerance by $8192\epsilon/8193$ gives the standard $(\epsilon, \sqrt{M}\epsilon)$ guarantee without changing any complexity order.

Lemma 3.1 (Cubic path budget). *Fix $\epsilon > 0$ and suppose that [Assumptions 2.1](#) and [3.1](#) hold and $\inf_x F(x) > -\infty$, and write $\Delta_F := F(x_0) - \inf_x F(x)$. Let K and $r_k = \|d_k\|$ be generated by [\(3.3\)](#)–[\(3.4\)](#), where $K < \infty$ is the return index guaranteed by [Theorem 3.1](#). Then*

$$M \sum_{k=0}^{K-1} r_k^3 \leq 4\Delta_F + \frac{K}{98\,304\sqrt{6}} \frac{\epsilon^{3/2}}{\sqrt{M}}. \quad (3.7)$$

4 Certified-inexact adaptive clipped UTR

The following adaptive method does not require the value of M as an algorithmic input. It accepts certified approximations of the objective, gradient, and Hessian; exact oracles are the zero-error special case. This formulation is essential for the NC–SC specialization in [Section 5](#), where the value function is not evaluated exactly. The oracle cost required to produce a requested certificate is kept separate from the number of certified trials.

Remark 4.1 (Why the fixed-scale estimate remains separate). *The fixed-scale result in [Section 3](#) and the adaptive result below share the trust-region geometry, but the former is not a formal special case of the latter. The fixed-scale procedure accepts every step without querying function values, permits a controlled increase on an inexact interior step, and supplies the cubic-path budget [\(3.7\)](#). The adaptive method instead filters every step through certified value and gradient tests and uses a padded Hessian model. Consequently, specializing the adaptive theorem would neither justify the unconditional fixed-scale update nor recover its cubic-path estimate. We keep the two counting arguments separate, but derive both one-step bounds from the common estimate in [Lemma A.1](#); that estimate, in turn, uses the single trust-region characterization in [Lemma 2.2](#).*

Assumption 4.1 (Certified adaptive oracle and parameter regime). *Suppose [Assumption 2.1](#) holds and $F^* > -\infty$. Fix*

$$\epsilon > 0, \quad 0 < \sigma_{\min} \leq \sigma_0, \quad \gamma_+ > 1, \quad \gamma_- > 1, \quad 0 < \eta \leq \frac{1}{256},$$

together with a value radius $\omega \geq 0$ and a gradient radius $0 \leq \delta_g \leq \epsilon/256$. Define

$$\sigma_* := \sqrt{\frac{M}{6}}, \quad \bar{\sigma} := \max\{\sigma_0, \gamma_+ \sigma_*\}, \quad (4.1)$$

and assume

$$\frac{\delta_g^2}{\sigma_* \sqrt{\epsilon}} + \frac{\omega}{4} \leq \omega, \quad (4.2a)$$

$$\left(\frac{15}{1024} - \eta \right) \frac{\epsilon^{3/2}}{\bar{\sigma}} \geq \frac{5\omega}{4}. \quad (4.2b)$$

The associated dimensionless inexactness load satisfies

$$\chi := \left(1 + \frac{4}{\log 2} \right) \frac{\bar{\sigma}\omega}{\eta\epsilon^{3/2}} < 1. \quad (4.3)$$

At a current iterate x_k , a value-gradient query returns a stored pair $(\tilde{F}_k, g_k)$ satisfying

$$|\tilde{F}_k - F(x_k)| \leq \frac{\omega}{8}, \quad \|g_k - \nabla F(x_k)\| \leq \delta_g. \quad (4.4)$$

At a candidate $x_k^+ = x_k + d$, the notation $(\tilde{F}_k^+, g_k^+)$ denotes a pair satisfying the same bounds with x_k replaced by x_k^+ . For every trial scale $\sigma > 0$, a Hessian query at x_k returns a symmetric $H_{k,\sigma}$ satisfying

$$\|H_{k,\sigma} - \nabla^2 F(x_k)\| \leq \frac{\sigma\sqrt{\epsilon}}{64}. \quad (4.5)$$

An accepted candidate value-gradient pair is stored without recomputation. A new Hessian may be requested at the next trial scale.

At an outer iterate x_k , write

$$h_k := \max\{\|g_k\|, \epsilon\}. \quad (4.6)$$

For a trial value $\sigma > 0$, consider

$$\begin{aligned} \min_{d \in \mathbb{R}^n} \quad & g_k^T d + \frac{1}{2} d^T \left(H_{k,\sigma} + \frac{\sigma\sqrt{\epsilon}}{64} I + \sigma\sqrt{h_k} I \right) d, \\ \text{s. t.} \quad & \|d\| \leq \frac{\sqrt{h_k}}{4\sigma}. \end{aligned} \quad (4.7)$$

The subproblem is solved globally. Its trust-region multiplier is denoted by $\lambda \geq 0$, as supplied by [Lemma 2.2](#) with $B = H_{k,\sigma} + (\sigma\sqrt{\epsilon}/64 + \sigma\sqrt{h_k})I$ and $R = \sqrt{h_k}/(4\sigma)$. In particular, $\lambda > 0$ implies that the radius is active.

The trial is accepted if one of the following mutually exclusive tests holds:

$$\|d\| < \frac{\sqrt{h_k}}{4\sigma} : \quad \tilde{F}_k^+ \leq \tilde{F}_k + \omega, \quad \|g_k^+\| \leq \frac{1}{2}h_k; \quad (4.8a)$$

$$\|d\| = \frac{\sqrt{h_k}}{4\sigma} : \quad \tilde{F}_k - \tilde{F}_k^+ \geq \eta \left(\frac{h_k^{3/2}}{\sigma} + \lambda\|d\|^2 \right) + \omega, \quad (4.8b)$$

$$\|g_k^+\| \leq \frac{1}{2}h_k + \lambda\|d\|. \quad (4.8c)$$

Algorithm 1: Certified-inexact adaptive clipped UTR

Input: x_0 and parameters and certified oracle responses satisfying [Assumption 4.1](#)

```
1  $k \leftarrow 0, \sigma \leftarrow \sigma_0$ 
2 obtain and store  $(\tilde{F}_k, g_k)$  at  $x_0$ 
3 while true do
4    $h_k \leftarrow \max\{\|g_k\|, \epsilon\}$ 
5   repeat
6     obtain  $H_{k,\sigma}$  at  $(x_k, \sigma)$  satisfying (4.5)
7     if  $\|g_k\| \leq \epsilon$  and  $\lambda_{\min}(H_{k,\sigma}) \geq -(9/8)\sigma\sqrt{\epsilon}$  then
8       return  $x_k$ 
9     end
10    Solve (4.7) and obtain  $(d, \lambda)$ 
11    obtain  $(\tilde{F}_k^+, g_k^+)$  at  $x_k + d$ 
12    if (4.8) holds then
13       $x_{k+1} \leftarrow x_k + d$ 
14       $(\tilde{F}_{k+1}, g_{k+1}) \leftarrow (\tilde{F}_k^+, g_k^+)$ 
15       $\sigma \leftarrow \max\{\sigma_{\min}, \sigma/\gamma_-\}$ 
16       $k \leftarrow k + 1$ 
17      accept the step
18    else
19      discard  $(\tilde{F}_k^+, g_k^+)$ 
20       $\sigma \leftarrow \gamma_+\sigma$ 
21    end
22  until a step is accepted
23 end
```

The stopping test in Algorithm 1 is evaluated for every trial value of σ , including immediately after each rejected trial increases σ .

4.1 Trial-step estimates

The safe-scale, parameter-bound, and finite-prefix counting lemmas used to prove the main theorem are stated and proved in [Section A](#). The supporting estimates in [Section A](#) establish finite termination and the following main complexity guarantee.

Remark 4.2 (Parameter freedom versus certified inexactness). *The compatibility conditions in [Assumption 4.1](#) are analytical sufficient conditions, not tests performed by the algorithm. For a nonzero certificate radius, their verification may use M through σ_* and $\bar{\sigma}$. Thus the certified-inexact theorem is conditional on an oracle provider that can supply the stated radii; it does not claim that producing such certificates is parameter free. This does not affect the first contribution: with exact oracle responses, $\omega = \delta_g = 0$, all compatibility conditions hold automatically and the executable scale update never uses M .*

Theorem 4.1 (Certified-inexact adaptive UTR complexity). *Suppose [Assumption 4.1](#) holds and define $\Delta_F := F(x_0) - F^*$. Then [Algorithm 1](#) terminates after finitely many trials. If Q_ϵ denotes the total number of solved trial subproblems, then*

$$Q_\epsilon = O\left(\frac{\bar{\sigma}(\Delta_F + \omega)\epsilon^{-3/2} + \log_+(h_0/\epsilon) + 1}{1 - \chi} + \log_+\frac{\bar{\sigma}}{\sigma_0}\right), \quad (4.9)$$

where the hidden constant depends only on η and $\gamma_{\pm}$. The returned point $\hat{x}$ satisfies

$$\|\nabla F(\hat{x})\| \leq \epsilon + \delta_g, \quad \lambda_{\min}(\nabla^2 F(\hat{x})) \geq -\frac{73}{64}\bar{\sigma}\sqrt{\epsilon}. \quad (4.10)$$

Remark 4.3 (Exact-oracle specialization and the initial scale). Suppose $F^* > -\infty$ and [Assumption 2.1](#) holds, and run [Algorithm 1](#) with exact function values, gradients, and Hessians. Then $\omega = \delta_g = 0$, so the compatibility conditions in [Assumption 4.1](#) hold automatically. No restriction on σ_0 is needed for the general bound

$$Q_{\epsilon} = O\left(\bar{\sigma}\Delta_F\epsilon^{-3/2} + \log_+(h_0/\epsilon) + \log_+(\bar{\sigma}/\sigma_0) + 1\right), \quad \bar{\sigma} = \max\{\sigma_0, \gamma_+\sqrt{M/6}\}, \quad (4.11)$$

and the return guarantee is $\lambda_{\min}(\nabla^2 F(\hat{x})) \geq -(73/64)\bar{\sigma}\sqrt{\epsilon}$. The additional condition $\sigma_0 = O(\sqrt{M})$ is needed only when one wants to simplify $\bar{\sigma} = O(\sqrt{M})$ and obtain the sharp leading term $O(\sqrt{M}\Delta_F\epsilon^{-3/2})$ together with an $O(\sqrt{M}\epsilon)$ second-order guarantee. Without this initial-scale condition, the general bound and the $O(\bar{\sigma}\sqrt{\epsilon})$ curvature guarantee above remain valid.

The theorem counts certified trials and globally solved trust-region subproblems. In the exact-oracle specialization, the Hessian is evaluated once per trial and the candidate function and gradient once per solved subproblem. For an inexact implementation, the work needed to produce the certificates in [Assumption 4.1](#) must be counted separately. The persistent parameter update in [Algorithm 1](#) is the same amortization mechanism used in adaptive regularization methods [8]; the multiplier-aware acceptance test is what couples gradient growth to objective decrease.

5 NC–SC minimax optimization

5.1 Problem setting and value-function geometry

We first use the fixed-scale estimates of [Section 3](#) for

$$\min_{x \in \mathbb{R}^{d_x}} P(x), \quad P(x) := \max_{y \in \mathbb{R}^{d_y}} f(x, y). \quad (5.1)$$

The certified-inexact adaptive result in [Theorem 4.1](#) is also directly applicable once the corrected value and derivative certificates below are supplied. We retain the fixed-scale wrapper for the explicit total inner-work bound, because its cubic-path estimate is the mechanism that removes the leading $\log(1/\epsilon)$.

The minimax development has three layers. This section first derives the value-function geometry and a corrected oracle, and then analyzes a known-parameter outer method whose maximization routine is supplied as an abstract residual module. [Section 6](#) removes the parameters by using uniformly over-accurate SCAR calls, while [Section 7](#) combines a fixed-radius outer method with persistent SCAR to remove the resulting leading logarithms from the inner-oracle bound. The gradient-scaled variant in [Section 8](#) retains the latter bounds with a radius depending on the corrected gradient. Keeping the residual interface common to these constructions makes clear which conclusions use only correctness of the inner solve and which additionally use a particular work guarantee.

We impose the assumptions used by Luo et al. [4].

Assumption 5.1 (NC–SC regularity). The payoff f is twice continuously differentiable and satisfies:

- (i) ∇f is ℓ -Lipschitz on $\mathbb{R}^{d_x} \times \mathbb{R}^{d_y}$;
- (ii) $\nabla^2 f$ is ρ -Lipschitz on $\mathbb{R}^{d_x} \times \mathbb{R}^{d_y}$ for some $\rho > 0$;
- (iii) $f(x, \cdot)$ is μ -strongly concave for every x , where $0 < \mu \leq \ell$;

(iv) $P^* := \inf_x P(x) > -\infty$.

We write $\kappa := \ell/\mu$.

Strong concavity makes $y^*(x) := \arg \max_y f(x, y)$ unique and guarantees that $\nabla_{yy}^2 f(x, y)$ is nonsingular. Define the Schur complement

$$\mathcal{H}(x, y) := \nabla_{xx}^2 f(x, y) - \nabla_{xy}^2 f(x, y) (\nabla_{yy}^2 f(x, y))^{-1} \nabla_{yx}^2 f(x, y). \quad (5.2)$$

The main departure from Luo et al. [4] is the value-function gradient estimator. Their MCN method uses the raw plug-in gradient $\nabla_x f(x, y)$, whose error is linear in $\|y - y^*(x)\|$, together with the same Schur-complement Hessian (5.2). We instead use the Newton-corrected gradient introduced below, whose inner error is quadratic. This relaxation permits an inner accuracy of order $\sqrt{\epsilon}$ and an observable residual stopping rule; at the level of the proved upper bounds, it preserves the leading outer rate and removes the multiplicative $\log(1/\epsilon)$ from the leading inner-gradient complexity.

Lemma 5.1 (Value-function geometry). *Under Assumption 5.1, $y^*(\cdot)$ is κ -Lipschitz, P is twice continuously differentiable, and*

$$\nabla P(x) = \nabla_x f(x, y^*(x)), \quad (5.3a)$$

$$\nabla^2 P(x) = \mathcal{H}(x, y^*(x)). \quad (5.3b)$$

Moreover, ∇P is $(1 + \kappa)\ell$ -Lipschitz and

$$\|\nabla^2 P(x) - \nabla^2 P(x')\| \leq M_P \|x - x'\|, \quad M_P := 4\sqrt{2}\kappa^3\rho. \quad (5.4)$$

These properties follow from Luo et al. [4, Lemmas 1–3].

In view of (5.4), the target in this section is an $(\epsilon, \sqrt{M_P\epsilon})$ -second-order stationary point of P .

5.2 A residual-based inner maximization module

The outer method uses the inner maximization only through an observable residual interface. This separates outer correctness from the choice between AGD, SCAR, or another accelerated inner solver.

Lemma 5.2 (Residual-distance certificate). *Under Assumption 5.1, for every x and y ,*

$$\|y - y^*(x)\| \leq \frac{\|\nabla_y f(x, y)\|}{\mu} \leq \kappa \|y - y^*(x)\|. \quad (5.5)$$

Definition 5.1 (Inner residual module). *Given an outer point x , a warm start y^{in} , and a target $\tau > 0$, an inner residual module $\mathcal{S}$ returns*

$$y^{\text{out}} = \text{MaxRes}_{\mathcal{S}}(x, y^{\text{in}}, \tau) \quad \text{such that} \quad \|\nabla_y f(x, y^{\text{out}})\| \leq \tau. \quad (5.6)$$

The module may retain implementation-specific state between calls, but this state is hidden from the outer method. We call the module accelerated if there are implementation factors $C_{\mathcal{S}} > 0$ and $b_{\mathcal{S}} \geq 1$, which may depend on the implementation and on κ but not on x , τ , R_{in} , ϵ , or the call index, such that a call whose initial residual is $R_{\text{in}} := \|\nabla_y f(x, y^{\text{in}})\|$ uses at most

$$C_{\mathcal{S}}\sqrt{\kappa} \left[b_{\mathcal{S}} + \left\lceil \log_2 \max \left\{ 1, \frac{R_{\text{in}}}{\tau} \right\} \right\rceil \right] \quad (5.7)$$

evaluations of $\nabla_y f$. The terminal residual evaluation may be reused when constructing the corrected gradient.

The interface (5.6), rather than the work bound (5.7), is the only inner-solver property used by the outer convergence theorem. We next record AGD as one implementation. For a fixed x , starting from $y^{(0)}$, set $v^{(0)} = y^{(0)}$ and

$$\begin{aligned} y^{(j+1)} &= v^{(j)} + \frac{1}{\ell} \nabla_y f(x, v^{(j)}), \\ v^{(j+1)} &= y^{(j+1)} + \frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1} (y^{(j+1)} - y^{(j)}). \end{aligned} \quad (5.8)$$

Proposition 5.1 (AGD implementation). *Under Assumption 5.1, fix x , $y^{(0)}$, and $\tau > 0$. For $\kappa > 1$, the iterate in (5.8) satisfies*

$$\|y^{(N)} - y^*(x)\| \leq \sqrt{\kappa + 1} \left(1 - \frac{1}{\sqrt{\kappa}}\right)^{N/2} \|y^{(0)} - y^*(x)\|. \quad (5.9)$$

Consequently, after evaluating $R_{\text{in}} = \|\nabla_y f(x, y^{(0)})\|$, return immediately if $R_{\text{in}} \leq \tau$; otherwise run

$$N := \left\lceil 2\sqrt{\kappa} \log_+ \left(\frac{\sqrt{\kappa + 1} \kappa R_{\text{in}}}{\tau} \right) \right\rceil$$

AGD steps and verify the residual once at the terminal iterate. The initial gradient is reused in the first AGD step. This implements Definition 5.1 using at most

$$1 + \left\lceil 2\sqrt{\kappa} \log_+ \left(\frac{\sqrt{\kappa + 1} \kappa R_{\text{in}}}{\tau} \right) \right\rceil \quad (5.10)$$

inner-gradient evaluations. Hence it satisfies (5.7) with universal C_S and $b_S = O(1 + \log \kappa)$. When $\kappa = 1$, one gradient step is sufficient.

We count one evaluation of $\nabla_y f(x, y)$ as one inner-gradient oracle call. Corrected-gradient and Schur-complement constructions, trust-region subproblem solves, and any linear algebra used to apply $(\nabla_{yy}^2 f)^{-1}$ are reported separately.

5.3 Corrected oracle and minimax algorithm

We now construct certified value and derivative oracles for the value function from derivatives of f . At an approximate maximizer y , define the Newton-corrected value

$$\hat{P}(x, y) := f(x, y) - \frac{1}{2} \nabla_y f(x, y)^T (\nabla_{yy}^2 f(x, y))^{-1} \nabla_y f(x, y). \quad (5.11)$$

Since $\nabla_{yy}^2 f \prec 0$, the correction raises the raw payoff and is exact for a concave quadratic inner problem. The corrected gradient and Hessian estimators are

$$\hat{g}(x, y) := \nabla_x f(x, y) - \nabla_{xy}^2 f(x, y) (\nabla_{yy}^2 f(x, y))^{-1} \nabla_y f(x, y), \quad (5.12a)$$

$$\hat{H}(x, y) := \mathcal{H}(x, y). \quad (5.12b)$$

The gradient estimator is the oracle-level distinction from Luo et al. [4, Algorithm 2]: their method uses $\nabla_x f(x, y)$, whereas (5.12a) adds a Newton correction. Both methods use the Schur-complement Hessian $\mathcal{H}(x, y)$.

Lemma 5.3 (Corrected value and derivative errors). *Under Assumption 5.1, for every x and y ,*

$$|\hat{P}(x, y) - P(x)| \leq \frac{(3\kappa - 1)\rho}{12} \|y - y^*(x)\|^3 \leq \frac{\kappa\rho}{4} \|y - y^*(x)\|^3, \quad (5.13a)$$

$$\|\hat{g}(x, y) - \nabla P(x)\| \leq \frac{(1 + \kappa)\rho}{2} \|y - y^*(x)\|^2, \quad (5.13b)$$

$$\|\hat{H}(x, y) - \nabla^2 P(x)\| \leq (1 + \kappa)^2 \rho \|y - y^*(x)\|. \quad (5.13c)$$

Moreover, whenever $\|y - y^*(x)\| \leq \mu/\rho$, the Newton-corrected value satisfies the following κ -independent local bound:

$$|\hat{P}(x, y) - P(x)| \leq \frac{7\rho}{24} \|y - y^*(x)\|^3. \quad (5.14)$$

Set the outer residual target

$$\tau_{\text{out}} := \frac{\mu\sqrt{M_P\epsilon/6}}{64(1+\kappa)^2\rho}. \quad (5.15)$$

A further practical distinction from Luo et al. [4, Algorithm 2 and Theorem 2] is the inner termination rule. Their analysis prescribes an iteration count for each AGD solve, whereas our outer method asks the input module $\mathcal{S}$ to satisfy the directly evaluable condition $\|\nabla_y f(x, y)\| \leq \tau_{\text{out}}$. By (5.5), this guarantees $\|y - y^*(x)\| \leq \tau_{\text{out}}/\mu$. The resulting oracle certificates are recorded explicitly next.

Corollary 5.1 (Residual-to-oracle certificate). *Under Assumption 5.1, let $M_P = 4\sqrt{2}\kappa^3\rho$ and let τ_{out} be given by (5.15). Every point y satisfying $\|\nabla_y f(x, y)\| \leq \tau_{\text{out}}$ has the observable certificate*

$$\xi(x, y) := \frac{(1+\kappa)^2\rho}{\mu} \|\nabla_y f(x, y)\|, \quad (5.16)$$

and satisfies

$$\|\hat{g}(x, y) - \nabla P(x)\| < \frac{\epsilon}{8192}, \quad \|\hat{H}(x, y) - \nabla^2 P(x)\| \leq \xi(x, y) \leq \frac{1}{64} \sqrt{\frac{M_P}{6}} \sqrt{\epsilon}. \quad (5.17)$$

In particular, the fixed-scale inexact oracle conditions (3.1) hold with $F = P$ and $M = M_P$.

Thus the stopping decision requires neither $y^*(x)$ nor an a priori bound on its distance from the inner initializer. The target has the useful scale and lower bound

$$\frac{\tau_{\text{out}}}{\mu} = \Theta\left(\sqrt{\frac{\epsilon}{\kappa\rho}}\right) = \Theta\left(\kappa\sqrt{\frac{\epsilon}{M_P}}\right), \quad (5.18a)$$

$$\tau_{\text{out}} \geq \frac{\ell}{64\sqrt{3}} \sqrt{\frac{\epsilon}{M_P}}. \quad (5.18b)$$

The second inequality follows by substitution: the ratio of its left-hand side to its right-hand side is $4\kappa^2/(1+\kappa)^2 \geq 1$.

Algorithm 2: Certificate-sharp fixed-scale NC-SC UTR

Input: $x_0, y_{-1}, \epsilon > 0, \ell, \mu, \rho$, and an inner module $\mathcal{S}$ satisfying (5.6)

- 1 $\kappa \leftarrow \ell/\mu$, $M_P \leftarrow 4\sqrt{2}\kappa^3\rho$, and compute τ_{out} from (5.15)
- 2 $y_0 \leftarrow \text{MaxRes}_{\mathcal{S}}(x_0, y_{-1}, \tau_{\text{out}})$
- 3 **for** $k = 0, 1, \dots$ **do**
- 4 Form $g_k = \hat{g}(x_k, y_k)$ and $H_k = \hat{H}(x_k, y_k)$
- 5 $\xi_k \leftarrow (1+\kappa)^2\rho\|\nabla_y f(x_k, y_k)\|/\mu$
- 6 $h_k \leftarrow \max\{\|g_k\|, \epsilon\}$
- 7 **if** $\|g_k\| \leq \epsilon$ and $\lambda_{\min}(H_k) \geq -\sqrt{M_P\epsilon} + \xi_k$ **then**
- 8 **return** x_k
- 9 **end**
- 10 Solve (3.4) globally with $F = P$ and $M = M_P$, and obtain d_k
- 11 $x_{k+1} \leftarrow x_k + d_k$
- 12 $y_{k+1} \leftarrow \text{MaxRes}_{\mathcal{S}}(x_{k+1}, y_k, \tau_{\text{out}})$
- 13 **end**

The residual used to certify the inner solve is already available, so computing ξ_k requires no additional oracle call. In the worst case, (5.17) makes the curvature threshold at least as permissive as

$$-\left(\sqrt{6} - \frac{1}{64}\right) \sqrt{\frac{M_P}{6}} \sqrt{\epsilon};$$

when the inner module returns a smaller residual, the tested threshold moves further toward $-\sqrt{M_P \epsilon}$. Thus the test uses all of the curvature slack certified at the current point, rather than the factor-1/ $\sqrt{6}$ threshold $-\sqrt{M_P/6} \sqrt{\epsilon}$ used by the earlier conservative rule.

Thus the minimax algorithm never evaluates $P(x)$ and never solves the inner maximization to $O(\epsilon)$ distance. It uses only the residual contract (5.6), with a target whose associated distance is proportional to $\sqrt{\epsilon}$. The outer algorithm and its correctness proof are independent of how $\mathcal{S}$ attains that residual. The parameters in (5.15) and the outer scale M_P are still known in this baseline; using a parameter-free inner implementation alone does not make the full wrapper parameter-free.

5.4 Outer and inner complexity

Theorem 5.1 (NC-SC complexity under an inner residual module). *Suppose f satisfies Assumption 5.1 and define $P(x) = \max_y f(x, y)$. Fix (x_0, y_{-1}) and $\epsilon > 0$, and set*

$$\kappa = \ell/\mu, \quad M_P = 4\sqrt{2} \kappa^3 \rho, \quad \Delta_P = P(x_0) - \inf_x P(x).$$

Run Algorithm 2 with the residual target (5.15) and any inner module that returns a point with residual at most that target. Then the algorithm returns $\hat{x}$ satisfying

$$\|\nabla P(\hat{x})\| \leq \frac{8193}{8192} \epsilon, \quad \lambda_{\min}(\nabla^2 P(\hat{x})) \geq -\sqrt{M_P \epsilon}. \quad (5.19)$$

The numbers of outer iterations, corrected-gradient and Schur-complement constructions, and trust-region subproblem solves are bounded by

$$\boxed{O\left(\Delta_P \kappa^{3/2} \sqrt{\rho} \epsilon^{-3/2} + \log_+ \frac{\|\nabla P(x_0)\|}{\epsilon} + 1\right)}. \quad (5.20)$$

More precisely, the method performs K nonterminal outer iterations and trust-region subproblem solves and $K+1$ corrected-gradient and Schur-complement constructions. Here $y_0 := \text{MaxRes}_{\mathcal{S}}(x_0, y_{-1}, \tau_{\text{out}})$ is the first inner output, $h_0 := \max\{\|\hat{g}(x_0, y_0)\|, \epsilon\}$, and

$$K \leq 65 \sqrt{\frac{M_P}{6}} \Delta_P \epsilon^{-3/2} + 3 \log_+ \frac{h_0}{\epsilon} + 2. \quad (5.21)$$

If, in addition, a module call that starts with residual R uses at most $C_S \sqrt{\kappa} [b_S + \lceil \log_2 \max\{1, R/\tau_{\text{out}}\} \rceil]$ inner-gradient evaluations, where C_S and b_S are independent of ϵ and the call index, then the total number of inner-gradient evaluations is

$$\begin{aligned} & O\left(C_S \sqrt{\kappa} b_S \left[\Delta_P \sqrt{M_P} \epsilon^{-3/2} + \log_+ \frac{\|\nabla P(x_0)\|}{\epsilon} + 1\right]\right) \\ & + O\left(C_S \sqrt{\kappa} \log_+ \frac{\|\nabla_y f(x_0, y_{-1})\|}{\tau_{\text{out}}}\right). \end{aligned} \quad (5.22)$$

In particular, if $C_S = O(1)$ and $b_S = O(1 + \log \kappa)$, its leading term is

$$O\left((1 + \log \kappa) \Delta_P \kappa^2 \sqrt{\rho} \epsilon^{-3/2}\right), \quad (5.23)$$

with no multiplicative $\log(1/\epsilon)$. Replacing the internal tolerance by $8192\epsilon/8193$ gives the standard $(\epsilon, \sqrt{M_P \epsilon})$ guarantee without changing these orders.

Corollary 5.2 (AGD realization). *Choosing $\mathcal{S}$ to be the AGD module in [Proposition 5.1](#) satisfies the hypotheses of [Theorem 5.1](#) with $C_{\mathcal{S}} = O(1)$ and $b_{\mathcal{S}} = O(1 + \log \kappa)$. The resulting method returns the point in (5.19), has the outer complexity in (5.20), and uses $O((1 + \log \kappa)\Delta_P \kappa^2 \sqrt{\rho} \epsilon^{-3/2})$ inner-gradient evaluations in its leading term, where $\Delta_P = P(x_0) - \inf_x P(x)$. No $\log(1/\epsilon)$ multiplies that term. When $\kappa = 1$, the inner work is bounded by a numerical constant per call.*

Remark 5.1 (What is counted). *If the algorithm solves K trust-region subproblems, it has K nonterminal outer iterations and $K + 1$ corrected-gradient and Schur-complement constructions, including the terminal test. The matrix inversion or linear algebra used to apply $(\nabla_{yy}^2 f)^{-1}$ is not counted as an inner gradient evaluation. The final $\nabla_y f$ evaluation used by the residual certificate is reused in the corrected-gradient construction. If that inverse is implemented by Hessian-vector products, its complexity must be reported separately. For an accelerated implementation whose $C_{\mathcal{S}}$ and $b_{\mathcal{S}}$ are independent of ϵ , all $\log(1/\epsilon)$ dependence in (5.22) is additive: it occurs in the one-time initialization term and in $\log_+(\|\nabla P(x_0)\|/\epsilon)$. Neither logarithm multiplies $\Delta_P \sqrt{M_P} \epsilon^{-3/2}$.*

Remark 5.2 (Comparison with the raw-gradient analysis). *For the raw plug-in gradient $\nabla_x f(x, y)$ used by Luo et al. [4], smoothness gives*

$$\|\nabla_x f(x, y) - \nabla P(x)\| \leq \ell \|y - y^*(x)\|.$$

Thus an $O(\epsilon)$ outer-gradient error requires an $O(\epsilon/\ell)$ inner distance. After the summation simplification in Luo et al. [4, Theorem 2], the ϵ -dependent leading logarithm scales, up to numerical constants, as

$$\log\left(\kappa^{3/2} + \frac{\epsilon^{3/2}}{\rho^{3/2}\tilde{\epsilon}^3}\right), \quad \tilde{\epsilon} = \min\left\{\frac{\epsilon}{192\ell}, \frac{\sqrt{M_P}\epsilon}{48\rho}\right\}.$$

For sufficiently small ϵ , this is

$$\log\left(\kappa^{3/2} + \left(\frac{\ell}{\sqrt{\rho\epsilon}}\right)^3\right).$$

The quadratic error in (5.13b) changes the required inner distance to order $\sqrt{\epsilon}$. At the level of the two proved upper bounds, this removes the leading multiplicative $\log(1/\epsilon)$ while retaining the $1 + \log \kappa$ dependence.

5.5 Certified adaptive specialization

The same corrected tuple also supplies the value and derivative certificates required by the adaptive unconstrained theorem. We state the oracle reduction separately so that it does not interrupt the fixed-scale complexity argument above.

Corollary 5.3 (Certified adaptive NC-SC oracle). *Under [Assumption 5.1](#), let $\mathcal{S}$ satisfy the residual contract (5.6). Fix $\epsilon > 0$, $\omega > 0$, and $\delta_g > 0$, and define*

$$\delta_{VG} := \min\left\{\sqrt{\frac{2\delta_g}{(1+\kappa)\rho}}, \left(\frac{3\omega}{2(3\kappa-1)\rho}\right)^{1/3}\right\}, \quad (5.24a)$$

$$\delta_H(\sigma) := \frac{\sigma\sqrt{\epsilon}}{64(1+\kappa)^2\rho}. \quad (5.24b)$$

At a value-gradient query, call $\mathcal{S}$ from any warm start with target $\mu\delta_{VG}$ and return $(\hat{P}, \hat{g})$. At a Hessian query with trial scale σ , call it with target $\mu\delta_H(\sigma)$ and return $\hat{H}$; the two queries may use different inner points. The returned quantities satisfy

$$|\hat{P} - P| \leq \frac{\omega}{8}, \quad \|\hat{g} - \nabla P\| \leq \delta_g, \quad \|\hat{H} - \nabla^2 P\| \leq \frac{\sigma\sqrt{\epsilon}}{64}. \quad (5.25)$$

Consequently, set $M = M_P$ in [Assumption 4.1](#), take $\eta = 1/256$, and write

$$\bar{\sigma} := \max \left\{ \sigma_0, \gamma_+ \sqrt{M_P/6} \right\}.$$

The explicit choices

$$\omega := \frac{\eta \epsilon^{3/2}}{16\bar{\sigma}}, \quad \delta_g := \min \left\{ \frac{\epsilon}{256}, \sqrt{\frac{\omega \sqrt{M_P/6} \sqrt{\epsilon}}{2}} \right\} \quad (5.26)$$

satisfy the compatibility conditions in that assumption. Hence [Theorem 4.1](#), together with [Corollary 5.3](#), gives the following bound on certified outer trials, where $\Delta_P := P(x_0) - P^*$:

$$O \left(\bar{\sigma} \Delta_P \epsilon^{-3/2} + \log_+ \frac{\|\nabla P(x_0)\|}{\epsilon} + \log_+ (\bar{\sigma}/\sigma_0) + 1 \right). \quad (5.27)$$

The returned point satisfies

$$\|\nabla P(\hat{x})\| \leq \frac{257}{256} \epsilon, \quad \lambda_{\min}(\nabla^2 P(\hat{x})) \geq -\frac{73}{64} \bar{\sigma} \sqrt{\epsilon}. \quad (5.28)$$

This statement does not count the refinement work needed to produce the scale-dependent certificates.

Remark 5.3 (Scope of the result). *The minimax analysis uses the valid upper bound M_P to fix the trust-region scale. Thus (5.20) is a genuine bound on all solved subproblems. The certified adaptive theorem is also applicable through [Corollary 5.3](#); unlike the earlier exact-oracle formulation, it explicitly accounts for corrected value, gradient, and Hessian errors. That corollary bounds certified outer trials. It does not assert the same total inner-gradient bound, because producing its scale-dependent value certificates may require additional refinement. The fixed-scale result is therefore retained for the sharp, fully accounted inner-work estimate in (5.22).*

6 Parameter-free residual-module UTR with SCAR

The baseline wrapper in [Algorithm 2](#) uses the problem parameters in both τ_{out} and M_P ; its AGD realization also uses ℓ and μ internally. Replacing AGD by SCAR instantiates the same residual interface without inner curvature parameters, but does not remove the parameters from the outer wrapper. This section also changes the outer tolerances and scale update, thereby removing both uses. The resulting outer wrapper takes no problem parameter as input other than the requested tolerance ϵ ; with a parameter-free residual module, the same is true of the complete method. There is, however, an important quantifier in the theorem. After fixing common constants (ℓ, μ, ρ) , one unknown threshold $\epsilon_0(\ell, \mu, \rho) > 0$ works simultaneously for every payoff in the corresponding class and every initialization. The method is therefore class-uniform asymptotically parameter-free, but the algorithm cannot verify that $\epsilon \leq \epsilon_0$. The distinction between this result and a uniform residual-only certificate without a positive common lower bound on μ is detailed in [Section B.1](#).

6.1 SCAR as a residual-module realization

We first record the precise external result needed below. It is stated for minimization so that it applies to $q_x(y) := -f(x, y)$.

Theorem 6.1 (SCAR; Lan et al. [9]). *Let $\phi : \mathbb{R}^m \rightarrow \mathbb{R}$ be differentiable, μ -strongly convex, and have an L -Lipschitz gradient, where $0 < \mu \leq L$. Let $\tau > 0$ and $u_0 \in \mathbb{R}^m$. If $\|\nabla \phi(u_0)\| \leq \tau$, return u_0 . Otherwise, initialize Algorithm 5.1 of Lan et al. [9] with $\mu_0 \geq \mu$ and $0 < M_0 \leq 4L$. SCAR returns $\hat{u}$ such that*

$$\|\nabla \phi(\hat{u})\| \leq \tau. \quad (6.1)$$

The number of evaluations of $\nabla\phi$ is at most

$$(4 + 8\sqrt{5}C_1)\sqrt{\frac{L}{\mu}}\left[\left\lceil\log_4\frac{\mu_0}{\mu}\right\rceil + \left\lceil\log_2\frac{\|\nabla\phi(u_0)\|}{\tau}\right\rceil\right], \quad (6.2)$$

where C_1 is the universal constant in Theorem 4.1 of Lan et al. [9]. The line-search estimate returned by SCAR is at most $2L$. The cited theorem states the strict case $\mu < L$. Its proof uses an AR estimate that remains valid when $L = \mu$, so the same SCAR call and the displayed conclusions extend to this endpoint without a parameter-dependent branch.

SCAR can be initialized without knowing either curvature parameter.

Corollary 6.1 (SCAR realization of the residual module). *Under Assumption 5.1, fix x , an inner initial point y^{in} , and a target $\tau > 0$. Choose any $z \neq y^{\text{in}}$ and set*

$$a_x := \frac{\|\nabla_y f(x, z) - \nabla_y f(x, y^{\text{in}})\|}{\|z - y^{\text{in}}\|}. \quad (6.3)$$

Run SCAR on $q_x = -f(x, \cdot)$ with $\mu_0 = M_0 = a_x$ and residual tolerance τ . No value of ℓ , μ , or κ is supplied to the inner solver, and its output $\hat{y}$ satisfies

$$\|\nabla_y f(x, \hat{y})\| \leq \tau.$$

Including the secant initialization, its inner-gradient complexity is

$$O\left(\sqrt{\kappa}\left[1 + \log \kappa + \log_+ \frac{\|\nabla_y f(x, y^{\text{in}})\|}{\tau}\right]\right). \quad (6.4)$$

Consequently, secant-initialized SCAR implements Definition 5.1 with $C_S = O(1)$ and $b_S = O(1 + \log \kappa)$.

Thus the known-parameter outer wrapper Algorithm 2 may use either the AGD realization in Corollary 5.2 or the SCAR realization in Corollary 6.1; Theorem 5.1 applies to both without changing the outer proof. In that baseline wrapper the target τ_{out} and scale M_P are still parameter-dependent. The role of the present section is to change those outer quantities as well, not merely to exchange one implementation of MaxRes for another.

The published bound (6.2) counts gradient evaluations. SCAR is not function-value-free: its backtracking tests also evaluate ϕ . With a standard implementation using a constant number of function values per line-search trial, their number is of the same order as (6.4), up to an additive line-search logarithm. We keep the two oracle counts conceptually separate.

6.2 Uniform asymptotic accuracy

The parameter-free construction uses the corrected value (5.11) and its error bound from Lemma 5.3.

Throughout this section, assume $0 < \epsilon < 1$. Every inner call uses the residual tolerance ϵ^2 , and every nonmonotone value allowance is ϵ^3 .

Lemma 6.1 (Class-uniform asymptotic oracle accuracy). *Fix $\ell \geq \mu > 0$ and $\rho > 0$, and set $\kappa := \ell/\mu$ and $M_P := 4\sqrt{2}\kappa^3\rho$. There exists $\epsilon_{\text{or}} = \epsilon_{\text{or}}(\ell, \mu, \rho) > 0$ such that the following holds simultaneously for every payoff satisfying Assumption 5.1 with these common constants and every $0 < \epsilon \leq \epsilon_{\text{or}}$. For every x and y satisfying $\|\nabla_y f(x, y)\| \leq \epsilon^2$,*

$$\|\hat{g}(x, y) - \nabla P(x)\| \leq \frac{\epsilon}{256}, \quad (6.5a)$$

$$\|\hat{H}(x, y) - \nabla^2 P(x)\| \leq \frac{1}{64}\sqrt{\frac{M_P\epsilon}{6}}, \quad (6.5b)$$

$$|\hat{P}(x, y) - P(x)| \leq \frac{\epsilon^3}{8}. \quad (6.5c)$$

The same ϵ_{or} works uniformly over the stated parameter class.

6.3 A fully observable adaptive UTR scheme

At a current point, let $(\hat{P}, g, H)$ denote the corrected tuple computed after an inner-module call with tolerance ϵ^2 , and set $h = \max\{\|g\|, \epsilon\}$. For a trial scale $\sigma \geq \epsilon$, consider the padded quadratic subproblem

$$\min_{\|d\| \leq \sqrt{h}/(4\sigma)} g^T d + \frac{1}{2} d^T \left(H + \frac{\sigma\sqrt{\epsilon}}{64} I + \sigma\sqrt{h} I \right) d. \quad (6.6)$$

At every scale $\sigma^2 \geq M_P/6$, the Hessian padding dominates (6.5b) and makes the quadratic term an upper model. For each fixed parameter class, the actual gradient error is of the much smaller order $O(\epsilon^4)$; at a safe scale, its worst-case effect on an interior trial is only $O(\epsilon^{15/2})$ and is absorbed by the $O(\epsilon^3)$ nonmonotonicity. Thus the inner solve need not be refined to an accuracy depending on the possibly arbitrarily small trial step.

For a global solution $d \neq 0$, write $r = \|d\|$ and let $\lambda \geq 0$ be the multiplier of the Euclidean ball. Applying Lemma 2.2 uses $B = H + (\sigma\sqrt{\epsilon}/64 + \sigma\sqrt{h})I$ and $R = \sqrt{h}/(4\sigma)$.

Algorithm 3: Asymptotically parameter-free outer residual-module UTR

Input: x_0, y_{-1} , $0 < \epsilon < 1$, and an inner module $\mathcal{S}$ satisfying (5.6)

```

1  $x \leftarrow x_0$ ,  $\sigma \leftarrow \epsilon$ , and  $\eta \leftarrow 1/256$ 
2  $y \leftarrow \text{MaxRes}_{\mathcal{S}}(x_0, y_{-1}, \epsilon^2)$ 
3 compute  $(\hat{P}, g, H)$  from (5.11) and (5.12)
4 while true do
5    $h \leftarrow \max\{\|g\|, \epsilon\}$ 
6   if  $\|g\| \leq \epsilon$  and  $\lambda_{\min}(H) \geq -(9/8)\sigma\sqrt{\epsilon}$  then
7     return  $x$ 
8   end
9   Solve (6.6) globally and obtain  $(d, \lambda)$ ; set  $r \leftarrow \|d\|$ 
10  if  $r = 0$  then
11    return  $x$ 
12  end
13   $x^+ \leftarrow x + d$ 
14   $y^+ \leftarrow \text{MaxRes}_{\mathcal{S}}(x^+, y, \epsilon^2)$ 
15  compute  $(\hat{P}^+, g^+, H^+)$ 
16  if  $r < \sqrt{h}/(4\sigma)$ ,  $\hat{P}^+ \leq \hat{P} + \epsilon^3$ , and  $\|g^+\| \leq h/2$  then
17    accept the trial
18  else
19    if  $r = \sqrt{h}/(4\sigma)$ ,  $\hat{P} - \hat{P}^+ \geq \eta(h^{3/2}/\sigma + \lambda r^2) + \epsilon^3$ , and  $\|g^+\| \leq h/2 + \lambda r$  then
20      accept the trial
21    else
22      reject the trial
23    end
24  end
25  if the trial is accepted then
26     $(x, y, \hat{P}, g, H) \leftarrow (x^+, y^+, \hat{P}^+, g^+, H^+)$ 
27     $\sigma \leftarrow \max\{\epsilon, \sigma/2\}$ 
28  else
29    discard the candidate tuple and set  $\sigma \leftarrow 2\sigma$ 
30  end
31 end

```

Every quantity tested in Algorithm 3 is observable. In particular, if $\mathcal{S}$ itself is parameter-free, the algorithm never uses ℓ , μ , ρ , κ , M_P , or Δ_P . The residual target ϵ^2 is independent of σ , so a rejection changes only the robust trust-region model; the corrected tuple at the current point remains frozen. Secant-initialized SCAR from Corollary 6.1 is the implementation used for the explicit inner-oracle count below.

6.4 Theory of the parameter-free variant

We next prove the trial estimate that drives the complete complexity argument.

Lemma 6.2 (Safe-scale robust trial). *Under Assumption 5.1, fix $0 < \epsilon < 1$. Consider a nonzero trial generated by Algorithm 3; write $M_P := 4\sqrt{2}\kappa^3\rho$, $h := \max\{\|g\|, \epsilon\}$, $r := \|d\|$, and let λ be its trust-region multiplier. Suppose the current and candidate inner points both satisfy the module residual bound $\|\nabla_y f\| \leq \epsilon^2$, and suppose ϵ is small enough that (6.5) holds. If*

$$\sigma^2 \geq M_P/6, \tag{6.7}$$

then a nonzero trial step satisfies

$$P(x) - P(x+d) \geq \frac{1}{4}\sigma\sqrt{h}r^2 + \frac{1}{2}\lambda r^2 - \frac{(1+\kappa)\rho}{2\mu^2}\epsilon^4 r, \quad (6.8a)$$

$$\|g^+\| \leq \frac{117}{256}h + \lambda r. \quad (6.8b)$$

If $r = \sqrt{h}/(4\sigma)$, then also

$$P(x) - P(x+d) \geq \frac{h^{3/2}}{128\sigma} + \frac{1}{4}\lambda r^2 \quad (6.9)$$

without any additional smallness condition.

Lemma 6.3 (Accepted-step accounting). *Under [Assumption 5.1](#), fix $0 < \epsilon < 1$ and consider any finite prefix of N accepted nonterminal steps generated by [Algorithm 3](#). Assume (6.5) holds at every tuple in the prefix and all its trial scales are bounded by some $\bar{\sigma} > 0$. If*

$$\frac{5\bar{\sigma}\epsilon^{3/2}}{4\eta} \left(1 + \frac{4}{\log 2}\right) \leq \frac{1}{2}, \quad (6.10)$$

then

$$N = O\left(\bar{\sigma}\Delta_P\epsilon^{-3/2} + \log_+ \frac{\|g_0\|}{\epsilon} + 1\right), \quad (6.11)$$

where $\Delta_P := P(x_0) - P^*$, g_0 is the first stored corrected gradient, and the hidden constant is numerical. In particular, the right-hand side is independent of the prefix length N .

Theorem 6.2 (Class-uniform parameter-free outer NC-SC complexity). *Fix $\ell \geq \mu > 0$ and $\rho > 0$. There exists $\epsilon_0 = \epsilon_0(\ell, \mu, \rho) \in (0, 1)$ with the following property. Let f be any payoff satisfying [Assumption 5.1](#) with these common constants and define $P(x) = \max_y f(x, y)$. For every initialization (x_0, y_{-1}) , set*

$$\kappa = \ell/\mu, \quad M_P = 4\sqrt{2}\kappa^3\rho, \quad \Delta_P = P(x_0) - \inf_x P(x).$$

For every $0 < \epsilon \leq \epsilon_0$ and every residual module $\mathcal{S}$ satisfying (5.6), [Algorithm 3](#) terminates and its output $\hat{x}$ satisfies

$$\|\nabla P(\hat{x})\| \leq \frac{257}{256}\epsilon, \quad \lambda_{\min}(\nabla^2 P(\hat{x})) \geq -\sqrt{M_P\epsilon}. \quad (6.12)$$

Consequently, running the algorithm with internal tolerance $256\epsilon/257$ returns an $(\epsilon, \sqrt{M_P\epsilon})$ -second-order stationary point. The total number Q of solved robust trust-region subproblems obeys

$$Q = O\left(\Delta_P\sqrt{M_P}\epsilon^{-3/2} + \log_+ \frac{\|\nabla P(x_0)\|}{\epsilon} + \log_+ \frac{\sqrt{M_P}}{\epsilon} + 1\right). \quad (6.13)$$

Its leading problem-parameter dependence is therefore $O(\Delta_P\kappa^{3/2}\sqrt{\rho}\epsilon^{-3/2})$. If the module calls use none of $\ell, \mu, \rho, \kappa, M_P$, or Δ_P , then the complete executable method uses only ϵ and universal numerical constants.

Corollary 6.2 (SCAR inner-oracle complexity). *Under the hypotheses of [Theorem 6.2](#), choose $\mathcal{S}$ to be the secant-initialized SCAR module in [Corollary 6.1](#). If Q denotes the number of robust trust-region subproblems in (6.13), set*

$$G_Q := 1 + \sqrt{2(1+\kappa)\ell\left(\Delta_P + \frac{5}{4}Q\epsilon^3\right)}. \quad (6.14)$$

Then the total number N_y of evaluations of $\nabla_y f$ is

$$\begin{aligned} N_y = & O\left(\sqrt{\kappa}(Q+1)\left[1 + \log \kappa + \log\left(1 + \frac{\ell\sqrt{G_Q}}{4\epsilon^3}\right)\right]\right) \\ & + O\left(\sqrt{\kappa}\left[1 + \log \kappa + \log_+ \frac{\|\nabla_y f(x_0, y_{-1})\|}{\epsilon^2}\right]\right). \end{aligned} \quad (6.15)$$

Thus the uniform SCAR solves in this construction introduce a leading $\log(1/\epsilon)$ in the inner-oracle count, although the outer bound in (6.13) has no such leading factor.

Remark 6.1 (Interpretation of the theorem). The onset in Theorem 6.2 is uniform over every class with common constants (ℓ, μ, ρ) and over all initializations. The same executable algorithm is used without knowing these constants, but it cannot observe whether $\epsilon \leq \epsilon_0(\ell, \mu, \rho)$. Thus class-uniform asymptotic correctness does not by itself provide an a posteriori finite-accuracy certificate. The scalar construction in Section B.1 explains why the threshold cannot be chosen independently of a positive common lower bound on μ .

The proof gives the explicit sufficient threshold in (A.20) and (A.24). If ℓ and ρ are fixed while $\mu \downarrow 0$, the displayed sufficient threshold is $\Theta(\mu)$. In particular, a sixth-order restriction $O(\mu^6)$ is not intrinsic: it would result from requiring the Hessian-oracle error to be smaller than $\sigma\sqrt{\epsilon}$ even at the smallest trial scale $\sigma = \epsilon$. The proof instead needs this comparison only once the scale is safe. The ϵ^3 value allowance is also needed to keep the accepted-step absorption condition linear in μ ; retaining an ϵ^2 allowance would give a sufficient $O(\mu^3)$ onset, which is still sharper than $O(\mu^6)$ but not linear.

7 Adaptive fixed-radius SCAR trust region

This section gives a second parameter-free construction. It combines the fixed-radius trust-region mechanism of Luenberger and Ye [7, Section 8.7] with a persistent version of SCAR. Persistent SCAR is a sequence-amortized realization of the same residual interface in Definition 5.1; its cross-call work guarantee is established separately in Lemma 7.1. In contrast with Algorithm 3, each ordinary inner update performs a fixed number of observable residual halvings instead of restarting SCAR to a prescribed power of ϵ . Consequently, neither $\log(1/\epsilon)$ nor $\log \kappa$ multiplies the leading inner-oracle bound. A high-accuracy SCAR call is used only when the trust-region solution is interior, and such validation calls occur only during the additive scale search.

As in Theorem 6.2, the result is class-uniform asymptotic. For each fixed triple (ℓ, μ, ρ) , one unknown threshold $\epsilon_0(\ell, \mu, \rho) > 0$ works for every payoff in the class and every initialization. The algorithm itself uses only ϵ and universal numerical constants; the precise quantifier and parameter list are stated in Theorem 7.1.

7.1 Persistent SCAR and the working oracle

We first extract from the proof of Theorem 6.1 a sequential form of SCAR. Only the curvature calibration and the line-search estimate are retained between calls; accelerated momentum is restarted whenever the objective changes.

Lemma 7.1 (Persistent SCAR halving). *Let $\{\phi_j\}_{j \geq 1}$ be any sequence of differentiable functions that are all μ -strongly convex and have L -Lipschitz gradients. Set $\kappa := L/\mu$. Initialize the strong-convexity guess ν_0 and line-search estimate $\mathcal{M}_0$ so that*

$$\mu \leq \nu_0 \leq L, \quad 0 < \mathcal{M}_0 \leq 4L.$$

A persistent halving call at (ϕ_j, u) freezes $R = \|\nabla \phi_j(u)\|$ and runs one AR call used in Algorithm 5.1 of Lan et al. [9]. If the call returns v with $\|\nabla \phi_j(v)\| \leq R/2$, the call accepts v and retains the current curvature estimates. Otherwise it discards v , replaces ν by $\nu/4$, retains the updated line-search estimate, and repeats from u on the same function. If $R = 0$, the call returns u immediately and is counted as a trivial successful halving.

The same AR call is used at the endpoint $L = \mu$. The estimate from Theorem 4.1 of Lan et al. [9] used below remains valid there, so this endpoint introduces no branch depending on unknown curvature parameters.

Across any N successful halving calls, the total number of failed curvature guesses is at most

$$\left\lceil \log_4 \frac{\nu_0}{\mu} \right\rceil \leq \lceil \log_4 \kappa \rceil. \quad (7.1)$$

The total number of gradient evaluations is at most

$$(4 + 2\sqrt{10} C_1) \sqrt{\kappa} \left(N + \left\lceil \log_4 \frac{\nu_0}{\mu} \right\rceil \right), \quad (7.2)$$

where C_1 is the constant in Theorem 4.1 of Lan et al. [9], also used in (6.2). After any nontrivial AR call, the retained line-search estimate is at most $2L$; before the first such call, it remains at its admissible initial value at most $4L$.

For the rest of this section, define

$$\mathcal{A}_\epsilon := \log(e/\epsilon), \quad \sigma_0 := \mathcal{A}_\epsilon^{-1}, \quad c_0 := 2^{-12}, \quad R(\sigma) := \frac{\sqrt{\epsilon}}{4\sigma}, \quad (7.3)$$

and let

$$N_w := \left\lceil \log_2 \frac{1 + c_0}{c_0} \right\rceil. \quad (7.4)$$

A corrected tuple $(\hat{P}, \hat{g}, \hat{H})$ at (x, y) and scale σ is called *working-accurate* if

$$\|\nabla_y f(x, y)\| \leq c_0 \ell R(\sigma). \quad (7.5)$$

This condition is used only in the analysis; the algorithm maintains it by a fixed number of observable residual halvings and never uses ℓ .

Lemma 7.2 (Working residual invariant). *Under Assumption 5.1, fix $0 < \epsilon < 1$ and $\sigma > 0$. Suppose (7.5) holds at (x, y) . If $\|d\| \leq R(\sigma)$, then N_w persistent halving calls on $q_{x+d} = -f(x + d, \cdot)$, starting from y , produce y^+ satisfying*

$$\|\nabla_y f(x + d, y^+)\| \leq c_0 \ell R(\sigma).$$

If the scale is doubled at the same x , one persistent halving call produces a point satisfying (7.5) with scale 2σ .

The invariant implies

$$\|y - y^*(x)\| \leq c_0 \kappa R(\sigma) = \frac{c_0 \kappa \sqrt{\epsilon}}{4\sigma}. \quad (7.6)$$

For later accounting, introduce the analytical, nonobservable error majorant

$$\mathcal{E}_\epsilon(\sigma) := \frac{7\rho}{24} (c_0 \kappa R(\sigma))^3 = \frac{7c_0^3 \rho \kappa^3}{1536} \frac{\epsilon^{3/2}}{\sigma^3}. \quad (7.7)$$

Lemma 7.3 (Class-uniform working and validation accuracy). *Fix $\ell \geq \mu > 0$ and $\rho > 0$, let $\kappa = \ell/\mu$ and $M_P = 4\sqrt{2}\kappa^3\rho$, and consider any payoff satisfying Assumption 5.1 with these common constants. There exist $\epsilon_1 = \epsilon_1(\ell, \mu, \rho) > 0$ and $\epsilon_2 = \epsilon_2(\ell, \mu, \rho) > 0$ such that the following statements hold uniformly over this parameter class.*

(i) *For every $0 < \epsilon \leq \epsilon_1$, every $\sigma \geq \mathcal{A}_\epsilon^{-1}$, and every working-accurate tuple,*

$$|\hat{P}(x, y) - P(x)| \leq \mathcal{E}_\epsilon(\sigma). \quad (7.8)$$

(ii) After decreasing ϵ_1 so that $\mathcal{A}_\epsilon^{-1} \leq \sqrt{M_P/6}$ for all $0 < \epsilon \leq \epsilon_1$, every working-accurate tuple with $\sigma^2 \geq M_P/6$ satisfies

$$\begin{aligned}\|\hat{g} - \nabla P(x)\| &\leq \epsilon/1024, \\ \|\hat{H} - \nabla^2 P(x)\| &\leq \sigma\sqrt{\epsilon}/64, \\ |\hat{P} - P(x)| &\leq \epsilon^{3/2}/(4096\sigma).\end{aligned}\tag{7.9}$$

(iii) Choose any secant a_x as in (6.3). If a SCAR validation call at x returns y_v satisfying

$$\|\nabla_y f(x, y_v)\| \leq c_0 a_x \epsilon^{3/2},\tag{7.10}$$

then, for all $0 < \epsilon \leq \epsilon_2$ and all $\sigma \geq \mathcal{A}_\epsilon^{-1}$,

$$\begin{aligned}\|\hat{g}(x, y_v) - \nabla P(x)\| &\leq \epsilon/1024, \\ \|\hat{H}(x, y_v) - \nabla^2 P(x)\| &\leq \sigma\sqrt{\epsilon}/32.\end{aligned}\tag{7.11}$$

The same ϵ_1 and ϵ_2 work independently of the payoff, x , the tuple, and the chosen secant within the stated class.

7.2 The adaptive fixed-radius algorithm

Given a working-accurate corrected tuple, solve

$$\min_{\|d\| \leq R(\sigma)} \hat{g}^T d + \frac{1}{2} d^T \left(\hat{H} + \frac{65}{64} \sigma \sqrt{\epsilon} I \right) d.\tag{7.12}$$

The radius is fixed for a given (ϵ, σ) and does not depend on $\|\hat{g}\|$. If d is a global solution and $\lambda \geq 0$ is its trust-region multiplier, then [Lemma 2.2](#) applies with $B = \hat{H} + (65/64)\sigma\sqrt{\epsilon}I$ and $R = R(\sigma)$.

Algorithm 4: Parameter-free adaptive fixed-radius TR with persistent SCAR

Input: x_0, y_{-1} and $0 < \epsilon < 1$

- 1 $\mathcal{A}_\epsilon \leftarrow \log(e/\epsilon)$, $\sigma \leftarrow \mathcal{A}_\epsilon^{-1}$, $c_0 \leftarrow 2^{-12}$, and $x \leftarrow x_0$
- 2 Choose $z_0 \neq y_{-1}$, compute the secant a_0 from (6.3), and initialize the persistent SCAR state with $\nu_0 = \mathcal{M}_0 = a_0$
- 3 At x_0 , apply persistent SCAR halving from y_{-1} until $\|\nabla_y f(x_0, y)\| \leq c_0 a_0 R(\sigma)$; compute the corrected tuple $(\hat{P}, \hat{g}, \hat{H})$
- 4 **while** *true* **do**
- 5 Solve (7.12) globally and obtain d ; set $x^+ \leftarrow x + d$
- 6 Starting from y , apply N_w successful persistent SCAR halvings to q_{x^+} and call the result y^+
- 7 **if** $\|d\| = R(\sigma)$ **then**
- 8 compute $\hat{P}^+$ at (x^+, y^+)
- 9 **if** $\hat{P} - \hat{P}^+ \geq \epsilon^{3/2}/(256\sigma)$ **then**
- 10 compute $(\hat{g}^+, \hat{H}^+)$ at (x^+, y^+)
- 11 $(x, y, \hat{P}, \hat{g}, \hat{H}) \leftarrow (x^+, y^+, \hat{P}^+, \hat{g}^+, \hat{H}^+)$
- 12 **else**
- 13 discard the candidate tuple, set $\sigma \leftarrow 2\sigma$, apply one successful persistent halving call at the unchanged x , replace y by the returned inner point, and recompute the current corrected tuple
- 14 **end**
- 15 **else**
- 16 Choose $z^+ \neq y^+$ and compute at x^+ the secant a^+ from (6.3)
- 17 Continue persistent SCAR halving at x^+ until $\|\nabla_y f(x^+, y_v)\| \leq c_0 a^+ \epsilon^{3/2}$; compute $(\hat{g}_v, \hat{H}_v)$
- 18 **if** $\|\hat{g}_v\| \leq \epsilon/2$ and $\lambda_{\min}(\hat{H}_v) \geq -(21/8)\sigma\sqrt{\epsilon}$ **then**
- 19 **return** x^+
- 20 **else**
- 21 discard the candidate and validator tuples, set $\sigma \leftarrow 2\sigma$, apply one successful persistent halving call at the unchanged x , replace y by the returned inner point, and recompute the current corrected tuple
- 22 **end**
- 23 **end**
- 24 **end**

Every operation in Algorithm 4 is observable. The secants are used only to initialize SCAR and to specify the high-accuracy validation residual. Strong monotonicity and smoothness give $\mu \leq a_0, a^+ \leq \ell$, but neither endpoint is known to the algorithm. After an accepted boundary step, the candidate corrected tuple is formed at the already computed inner point and stored without further refinement. After a failed trial, the candidate is discarded, the current point is unchanged, and the refreshed tuple at the doubled scale replaces the stored tuple. This bookkeeping is essential for the proxy telescope below. The scale never decreases. The persistent SCAR state $(\nu, \mathcal{M})$ is retained after every accepted trial, rejected trial, and failed validation; rejecting an outer candidate never rolls back this state. The initial residual evaluation is counted as the first gradient evaluation of the corresponding AR call, and the final residual gradient is reused when forming the corrected tuple.

The initialization satisfies the working invariant because $a_0 \leq \ell$. Thereafter, Lemma 7.2 proves inductively that every tuple used in (7.12) is working-accurate.

7.3 Safe scale, termination, and outer complexity

The following lemma is the fixed-radius analogue of the large-scale acceptance result for adaptive UTR.

Lemma 7.4 (Class-uniform safe fixed-radius trial). *Fix $\ell \geq \mu > 0$ and $\rho > 0$, set $\kappa = \ell/\mu$ and $M_P := 4\sqrt{2}\kappa^3\rho$. There exists $\epsilon_3 = \epsilon_3(\ell, \mu, \rho) > 0$ such that the following holds simultaneously for every payoff satisfying [Assumption 5.1](#) with these common constants and every $0 < \epsilon \leq \epsilon_3$. Consider any trial generated by [Algorithm 4](#), and suppose its current and candidate tuples are working-accurate and its validator, when invoked, satisfies (7.10). Let d be its global trust-region solution and let λ be the multiplier supplied by [Lemma 2.2](#) with $B = \hat{H} + (65/64)\sigma\sqrt{\epsilon}I$ and $R = R(\sigma)$. Assertions (i)–(ii) below hold when the trial scale satisfies*

$$\sigma^2 \geq M_P/6, \quad (7.13)$$

and assertion (iii) holds at every generated scale $\sigma \geq \mathcal{A}_\epsilon^{-1}$.

(i) If $\|d\| = R(\sigma)$, then

$$P(x) - P(x+d) \geq \frac{63}{4096} \frac{\epsilon^{3/2}}{\sigma} + \frac{\lambda}{2} R(\sigma)^2, \quad (7.14)$$

and the observable boundary acceptance test succeeds.

(ii) If $\|d\| < R(\sigma)$, then, at $x^+ = x + d$,

$$\begin{aligned} \|\nabla P(x^+)\| &\leq 457\epsilon/1024, \\ \lambda_{\min}(\nabla^2 P(x^+)) &\geq -(81/32)\sigma\sqrt{\epsilon}. \end{aligned} \quad (7.15)$$

The validation test succeeds.

(iii) At any scale, not necessarily a safe one, success of the validation test implies

$$\|\nabla P(x^+)\| < \epsilon, \quad \lambda_{\min}(\nabla^2 P(x^+)) \geq -(85/32)\sigma\sqrt{\epsilon}. \quad (7.16)$$

Theorem 7.1 (Class-uniform parameter-free fixed-radius complexity). *Fix $\ell \geq \mu > 0$ and $\rho > 0$. There exists $\epsilon_0 = \epsilon_0(\ell, \mu, \rho) \in (0, 1)$ with the following property. Let f be any payoff satisfying [Assumption 5.1](#) with these common constants, define $P(x) = \max_y f(x, y)$, and fix any initialization (x_0, y_{-1}) . Set*

$$\kappa = \ell/\mu, \quad M_P = 4\sqrt{2}\kappa^3\rho, \quad \Delta_P = P(x_0) - \inf_x P(x).$$

Then, for every $0 < \epsilon \leq \epsilon_0$, [Algorithm 4](#) terminates. The algorithm uses only ϵ and universal numerical constants; in particular, it does not use $\ell, \mu, \rho, \kappa, M_P$, or Δ_P . Its returned point x_{out} satisfies

$$\|\nabla P(x_{\text{out}})\| < \epsilon, \quad \lambda_{\min}(\nabla^2 P(x_{\text{out}})) \geq -\frac{85}{16\sqrt{6}}\sqrt{M_P\epsilon}. \quad (7.17)$$

If Q is the total number of solved trust-region subproblems and N_y is the total number of evaluations of $\nabla_y f$, including the secant evaluations, then

$$Q = O\left(\Delta_P \sqrt{M_P} \epsilon^{-3/2}\right) + O_{\text{fixed}}(\text{polylog}(1/\epsilon)), \quad (7.18a)$$

$$N_y = O\left(\Delta_P \sqrt{\kappa M_P} \epsilon^{-3/2}\right) + O_{\text{fixed}}(\sqrt{\kappa} \text{polylog}(1/\epsilon)). \quad (7.18b)$$

Here O_{fixed} may depend on the fixed payoff and initialization but not on ϵ . Equivalently, the leading outer and inner terms are, respectively,

$$O(\Delta_P \kappa^{3/2} \sqrt{\rho} \epsilon^{-3/2}) \quad \text{and} \quad O(\Delta_P \kappa^2 \sqrt{\rho} \epsilon^{-3/2}).$$

Thus neither $\log(1/\epsilon)$ nor $\log \kappa$ multiplies the leading inner-gradient term. Running the method with internal tolerance $(1536/7225)\epsilon$ gives the standard $(\epsilon, \sqrt{M_P\epsilon})$ guarantee, provided that internal tolerance lies in the class-uniform regime above.

Remark 7.1 (Scope of the fixed-radius guarantee). *The onset in Theorem 7.1 is uniform over the class with common constants (ℓ, μ, ρ) and over all initializations, but it is not observable to the parameter-free algorithm. The residual-only obstruction for prescribed absolute tolerances, its precise relation to the secant-scaled targets used here, and the roles of the corrected-value telescope and the interior validator are explained in Section B.2.*

Remark 7.2 (Oracle accounting). *The bound in Theorem 7.1 counts inner-gradient evaluations, which is the oracle counted in Theorem 6.1. SCAR’s line search also evaluates $q_x = -f(x, \cdot)$. Under the standard implementation in which the accelerated subroutine uses $O(1)$ function values per gradient evaluation, the function-evaluation remark following Theorem 4.1 of Lan et al. [9] bounds the extra cost of an AR call containing S_{AR} backtracking stages by $2S_{\text{AR}} + \log_2(\mathcal{M}^+/\mathcal{M})$. Summed over all persistent calls, the stage counts are absorbed by N_y , while the logarithmic ratios telescope because $\mathcal{M}$ is persistent. In particular,*

$$\log_2(\mathcal{M}_{\text{final}}/a_0) \leq \log_2(2\ell/a_0) \leq 1 + \log_2 \kappa.$$

Thus the total inner function-value count is $O(N_y + \log \kappa)$. The initial corrected tuple, trial candidates, and scale refreshes use at most $2Q + 1$ additional payoff values. Each stored tuple and each validator also constructs corrected gradient and Hessian estimators. The inverse actions in (5.11) and (5.12) are treated as exact linear algebra, as in the earlier minimax analysis; iterative linear solves require a separate accuracy count.

8 Adaptive gradient-scaled SCAR trust region

We next allow the trust-region radius to depend on the corrected gradient while retaining the working accuracy and persistent SCAR module of Section 7. The number of tracking halvings now depends on the step length. A gradient-level potential and a cubic-path estimate control their cumulative cost.

8.1 Working accuracy and tracking

Retain $\mathcal{A}_\epsilon$, σ_0 , and c_0 from (7.3), and distinguish the inner scale

$$r_\epsilon(\sigma) := \frac{\sqrt{\epsilon}}{4\sigma}$$

from the outer radius. A stored tuple is working-accurate when

$$\|\nabla_y f(x, y)\| \leq c_0 \ell r_\epsilon(\sigma). \quad (8.1)$$

This is the same invariant as (7.5); the unknown ℓ is used only in its analysis.

Lemma 8.1 (Variable-step tracking). *Suppose (8.1) holds at (x, y) , and let d be any step with $r = \|d\|$. Starting from y at $x + d$, apply*

$$N_{\text{tr}}(r, \sigma) := \left\lceil \log_2 \left(1 + \frac{r}{c_0 r_\epsilon(\sigma)} \right) \right\rceil \quad (8.2)$$

successful persistent SCAR halvings. The returned point satisfies (8.1) at $x + d$ with the same scale. At an unchanged x , one successful halving restores the invariant after σ is doubled.

The following error bounds will also be used at unsafe scales. Set

$$C_g := \frac{c_0^2(1 + \kappa)\rho\kappa^2}{32}, \quad C_v := \frac{7c_0^3\rho\kappa^3}{1536}. \quad (8.3)$$

Lemma 8.2 (Working-oracle errors). *Fix the common constants (ℓ, μ, ρ) in [Assumption 5.1](#). For all sufficiently small ϵ , uniformly over the class, every working-accurate tuple at $\sigma \geq \sigma_0$ satisfies*

$$\|\hat{g} - \nabla P(x)\| \leq C_g \frac{\epsilon}{\sigma^2}, \quad |\hat{P} - P(x)| \leq C_v \frac{\epsilon^{3/2}}{\sigma^3}. \quad (8.4)$$

The safe-scale bounds [\(7.9\)](#) and the validation bounds [\(7.11\)](#) remain valid.

8.2 The gradient-scaled algorithm

Given a working-accurate tuple, define

$$h := \max\{\|\hat{g}\|, \epsilon\}, \quad R(h, \sigma) := \frac{\sqrt{h}}{4\sigma}, \quad (8.5)$$

and solve

$$\min_{\|d\| \leq R(h, \sigma)} \hat{g}^T d + \frac{1}{2} d^T \left(\hat{H} + \frac{\sigma \sqrt{\epsilon}}{64} I + \sigma \sqrt{h} I \right) d. \quad (8.6)$$

For a global solution d and its multiplier $\lambda \geq 0$, write

$$r := \|d\|, \quad W := \sigma \sqrt{h} r^2, \quad U := \lambda r^2, \quad T := \frac{\epsilon^{3/2}}{\sigma}. \quad (8.7)$$

When $h = \epsilon$, the model and radius coincide with [\(7.12\)](#). Otherwise the larger radius is accompanied by the regularization $\sigma \sqrt{h} I$.

Algorithm 5: Parameter-free gradient-scaled TR with persistent SCAR

Input: x_0, y_{-1} and $0 < \epsilon < 1$

- 1 $\mathcal{A}_\epsilon \leftarrow \log(e/\epsilon)$, $\sigma \leftarrow \mathcal{A}_\epsilon^{-1}$, $c_0 \leftarrow 2^{-12}$, and $x \leftarrow x_0$
- 2 Choose $z_0 \neq y_{-1}$, compute a_0 from (6.3), and initialize the persistent SCAR state with $\nu_0 = \mathcal{M}_0 = a_0$
- 3 At x_0 , apply persistent SCAR halving from y_{-1} until $\|\nabla_y f(x_0, y)\| \leq c_0 a_0 r_\epsilon(\sigma)$; compute $(\hat{P}, \hat{g}, \hat{H})$
- 4 **while** *true* **do**
- 5 Set $h \leftarrow \max\{\|\hat{g}\|, \epsilon\}$, solve (8.6) globally, and obtain (d, λ)
- 6 Set $x^+ \leftarrow x + d$, $r \leftarrow \|d\|$, and form (W, U, T) from (8.7)
- 7 Starting from y , apply $N_{\text{tr}}(r, \sigma)$ successful persistent SCAR halvings to q_{x^+} and call the result y^+
- 8 **if** $h = \epsilon$ and $r < R(h, \sigma)$ **then**
- 9 Choose $z^+ \neq y^+$ and compute at x^+ the secant a^+ from (6.3)
- 10 Continue persistent SCAR halving at x^+ until $\|\nabla_y f(x^+, y_v)\| \leq c_0 a^+ \epsilon^{3/2}$; compute $(\hat{g}_v, \hat{H}_v)$
- 11 **if** $\|\hat{g}_v\| \leq \epsilon/2$ and $\lambda_{\min}(\hat{H}_v) \geq -(21/8)\sigma\sqrt{\epsilon}$ **then**
- 12 **return** x^+
- 13 **else**
- 14 discard the candidate and validator tuples, set $\sigma \leftarrow 2\sigma$, apply one successful persistent halving at the unchanged x , replace y by the returned point, and recompute the current corrected tuple
- 15 **end**
- 16 **else**
- 17 Compute $(\hat{P}^+, \hat{g}^+, \hat{H}^+)$ at (x^+, y^+)
- 18 **if** $\hat{P} - \hat{P}^+ \geq W/8 + U/4 - T/1024$ and $\|\hat{g}^+\| \leq h/2 + \lambda r$ **then**
- 19 $(x, y, \hat{P}, \hat{g}, \hat{H}) \leftarrow (x^+, y^+, \hat{P}^+, \hat{g}^+, \hat{H}^+)$
- 20 **else**
- 21 discard the candidate tuple, set $\sigma \leftarrow 2\sigma$, apply one successful persistent halving at the unchanged x , replace y by the returned point, and recompute the current corrected tuple
- 22 **end**
- 23 **end**
- 24 **end**

An accepted candidate tuple is stored without further refinement. The scale is nondecreasing, and the persistent state $(\nu, \mathcal{M})$ is retained after accepted trials, rejected trials, and failed validations. As in Section 7, only the curvature estimates are retained across changes of the inner objective; accelerated momentum is restarted. The initial residual query is counted as the first gradient evaluation of the corresponding AR call, and the terminal residual gradient is reused in the corrected tuple. Initialization establishes (8.1) because $a_0 \leq \ell$, and Lemma 8.1 preserves it.

8.3 Safe scale and complexity

Lemma 8.3 (Safe scale for the gradient-scaled method). *Under Assumption 5.1, for all sufficiently small ϵ with a threshold depending only on (ℓ, μ, ρ) , the following hold. If $\sigma^2 \geq M_P/6$, every ordinary trial of Algorithm 5 satisfies both acceptance tests, and every trial with $h = \epsilon$ and $r < R(h, \sigma)$ passes validation. At any scale $\sigma \geq \sigma_0$, successful validation implies*

$$\|\nabla P(x^+)\| < \epsilon, \quad \lambda_{\min}(\nabla^2 P(x^+)) \geq -\frac{85}{32}\sigma\sqrt{\epsilon}. \quad (8.8)$$

Theorem 8.1 (Class-uniform gradient-scaled complexity). *Fix $\ell \geq \mu > 0$ and $\rho > 0$. There exists $\epsilon_0 = \epsilon_0(\ell, \mu, \rho) \in (0, 1)$ such that the following holds for every payoff satisfying [Assumption 5.1](#) with these common constants and every initialization (x_0, y_{-1}) . Set $\kappa = \ell/\mu$, $M_P = 4\sqrt{2}\kappa^3\rho$, and $\Delta_P = P(x_0) - \inf_x P(x)$. For every $0 < \epsilon \leq \epsilon_0$, [Algorithm 5](#) terminates using only ϵ and universal numerical constants. Its output satisfies*

$$\|\nabla P(x_{\text{out}})\| < \epsilon, \quad \lambda_{\min}(\nabla^2 P(x_{\text{out}})) \geq -\frac{85}{16\sqrt{6}}\sqrt{M_P\epsilon}. \quad (8.9)$$

If Q counts all solved trust-region subproblems and N_y counts all inner-gradient evaluations, including the secant evaluations, then

$$Q = O\left(\Delta_P \sqrt{M_P} \epsilon^{-3/2}\right) + O_{\text{fixed}}(\mathcal{A}_\epsilon^3), \quad (8.10a)$$

$$N_y = O\left(\Delta_P \sqrt{\kappa M_P} \epsilon^{-3/2}\right) + O_{\text{fixed}}(\sqrt{\kappa} \mathcal{A}_\epsilon^3). \quad (8.10b)$$

Here O_{fixed} may depend on the fixed payoff, regularity constants, and initialization, but not on ϵ . The leading outer and inner terms are equivalently

$$O(\Delta_P \kappa^{3/2} \sqrt{\rho} \epsilon^{-3/2}) \quad \text{and} \quad O(\Delta_P \kappa^2 \sqrt{\rho} \epsilon^{-3/2}).$$

Using internal tolerance $(1536/7225)\epsilon$ gives the standard $(\epsilon, \sqrt{M_P\epsilon})$ guarantee whenever that internal tolerance is in the stated regime.

The fixed-radius method uses a constant number of working halvings per trial. Here the accepted-step analysis instead uses

$$\mathcal{L}_\epsilon(h) := \lceil \log_2(h/\epsilon) \rceil, \quad \Phi := \hat{P} + \frac{\epsilon^{3/2}}{512\sigma} \mathcal{L}_\epsilon(h). \quad (8.11)$$

Interior acceptance decreases the gradient level, while boundary acceptance pays for a possible level increase through λr^2 . After accounting for scale refreshes, the same budget controls $\sum \sigma_k^2 \|d_k\|^3$ and the total tracking work. The proofs are given in [Section A.7](#). The class-uniform interpretation and exact-linear-algebra convention are the same as in [Section 7](#).

9 Conclusion

For unconstrained optimization, the certified-inexact adaptive UTR removes M from the outer scale update and admits the value and derivative certificates needed by the minimax reduction. Producing nonzero certified radii remains an oracle-interface requirement. The exact-oracle specialization is fully parameter free and retains sharp leading dependence on Δ_F , M , and ϵ with only additive search overhead. The fixed-scale inexact analysis is used as an outer estimate rather than a separate algorithm; in particular, its cubic-path budget is what amortizes the inner work in the minimax specialization.

For NC–SC minimax optimization, the outer method now receives a residual-based maximization module as an input. Correctness depends only on the module’s returned residual, while AGD, restarted SCAR, and persistent SCAR provide different work guarantees for the same interface. In the fixed-scale baseline, that residual also provides the Hessian-error certificate used by the worst-case-sharp stopping test, so no curvature slack is lost when transferring the estimator test to the true value-function Hessian. The Newton-corrected value-function gradient permits inner solves at a $\sqrt{\epsilon}$ scale. This preserves the $O(\Delta_P \kappa^{3/2} \sqrt{\rho} \epsilon^{-3/2})$ outer complexity and removes the leading multiplicative $\log(1/\epsilon)$ from the raw-gradient tracking bound, while retaining logarithmic dependence on κ for the restarted accelerated implementations.

The first parameter-free residual-module UTR removes all problem parameters from the executable outer and inner algorithms and retains the sharp outer leading order uniformly over each class with common

constants (ℓ, μ, ρ) once ϵ is sufficiently small, but its uniformly accurate inner solves restore a leading $\log(1/\epsilon)$. The adaptive fixed-radius variant replaces those ordinary solves by persistent residual-halving calls. It preserves the same leading outer order and attains leading inner-gradient complexity $O(\Delta_P \kappa^2 \sqrt{\rho} \epsilon^{-3/2})$, with all $\log(1/\epsilon)$ and $\log \kappa$ terms confined to additive initialization, validation, and scale-search costs. The gradient-scaled variant retains these leading bounds; its variable tracking work is amortized by a gradient-level potential and a cubic-path budget. These parameter-free conclusions are asymptotic but class-uniform: their onset depends only on the common regularity constants and not on the payoff or initialization. For the first construction, the displayed sufficient onset is $\Theta(\mu)$ for fixed ℓ and ρ . Because the executable algorithms neither know nor certify the relevant onset, this is still weaker than an a posteriori finite-accuracy guarantee.

A Deferred technical proofs

The main text states the assumptions and results used by subsequent sections. This appendix collects the technical proofs in their order of appearance.

A.1 A common inexact trust-region estimate

Lemma A.1 (Common inexact trust-region step estimate). *Suppose [Assumption 2.1](#) holds. At a point x , let g and a symmetric H satisfy*

$$\|g - \nabla F(x)\| \leq \delta_0, \quad \|H - \nabla^2 F(x)\| \leq \xi.$$

For $a \geq 0$ and $R > 0$, let d be a global solution of

$$\min_{\|u\| \leq R} q(u) := g^T u + \frac{1}{2} u^T (H + aI) u,$$

let λ be the multiplier supplied by [Lemma 2.2](#), and write $r = \|d\|$. Let a candidate gradient g^+ satisfy $\|g^+ - \nabla F(x + d)\| \leq \delta_+$. Then

$$q(d) \leq -\frac{\lambda}{2} r^2, \tag{A.1a}$$

$$F(x) - F(x + d) \geq \frac{a - \xi}{2} r^2 + \frac{\lambda}{2} r^2 - \frac{M}{6} r^3 - \delta_0 r, \tag{A.1b}$$

$$\|g^+\| \leq (a + \xi + \lambda) r + \frac{M}{2} r^2 + \delta_0 + \delta_+. \tag{A.1c}$$

Proof. Apply (2.4) with $B = H + aI$. Its stationarity and positive-semidefiniteness conditions give

$$q(d) = -\frac{1}{2} d^T (H + aI) d - \lambda r^2 \leq -\frac{\lambda}{2} r^2,$$

which proves (A.1a). The Taylor bound and the current endpoint errors imply

$$F(x) - F(x + d) \geq -g^T d - \frac{1}{2} d^T H d - \delta_0 r - \frac{\xi}{2} r^2 - \frac{M}{6} r^3.$$

Since $-g^T d - d^T H d / 2 = -q(d) + a r^2 / 2$, the model bound yields (A.1b).

For the candidate gradient, decompose the two endpoint errors, Taylor remainder, Hessian error, and model residual:

$$\begin{aligned} g^+ &= [g^+ - \nabla F(x + d)] + [\nabla F(x + d) - \nabla F(x) - \nabla^2 F(x) d] \\ &\quad + [\nabla F(x) - g] + [\nabla^2 F(x) - H] d + [g + H d]. \end{aligned}$$

The stationarity part of (2.4), again with $B = H + aI$, gives $g + H d = -(a + \lambda) d$. Taking norms proves (A.1c). $\square$

A.2 Fixed-scale inexact UTR estimates

Let $K \in \mathbb{N}_0 \cup \{\infty\}$ be the first return index of the procedure (3.3)–(3.4), with $K = \infty$ if it never returns. For $k < K$, set $r_k := \|d_k\|$ and let λ_k be the multiplier supplied by Lemma 2.2 with

$$g = g_k, \quad B = H_k + \xi_k I + 2\sqrt{M/6}\sqrt{h_k} I, \quad R = \frac{\sqrt{h_k}}{4\sqrt{M/6}}.$$

Lemma A.2 (Decrease and clipped-gradient recurrence). *Fix $\epsilon > 0$ and suppose Assumptions 2.1 and 3.1 hold. Every nonterminal step $k < K$ generated by (3.3)–(3.4) satisfies*

$$F(x_k) - F(x_{k+1}) \geq \frac{3}{4}\sqrt{\frac{M}{6}}\sqrt{h_k}r_k^2 + \frac{1}{2}\lambda_k r_k^2 - \frac{\epsilon r_k}{8192}. \quad (\text{A.2})$$

Consequently,

$$\lambda_k = 0 : \quad F(x_k) - F(x_{k+1}) \geq -\frac{\epsilon^{3/2}}{201\,326\,592\sqrt{M/6}}, \quad (\text{A.3a})$$

$$\lambda_k > 0 : \quad F(x_k) - F(x_{k+1}) \geq \frac{1535}{32768} \frac{h_k^{3/2}}{\sqrt{M/6}} + \frac{1}{2}\lambda_k r_k^2. \quad (\text{A.3b})$$

Moreover,

$$h_{k+1} \leq \max \left\{ \frac{2849}{4096} h_k + \lambda_k r_k, \epsilon \right\}. \quad (\text{A.4})$$

The small negative term in (A.3a) is why a uniform $O(\epsilon)$ gradient error is admissible. An inexact step with $\lambda_k = 0$ need not be monotone: it may increase F by at most $\epsilon^{3/2}/(201\,326\,592\sqrt{M/6})$. The finite-prefix argument below debits every such increase against the decrease on indices in $\mathcal{B}_N$.

Proof of Lemma A.2. Apply Lemma A.1 with

$$a = \xi_k + 2\sqrt{M/6}\sqrt{h_k}, \quad R = \frac{\sqrt{h_k}}{4\sqrt{M/6}}, \quad \xi = \xi_k, \quad \delta_0 = \delta_+ = \frac{\epsilon}{8192}.$$

Because $r_k \leq R$, $(M/6)r_k^3 \leq \sqrt{M/6}\sqrt{h_k}r_k^2/4$. The certificate padding cancels the Hessian-error term in (A.1b); hence

$$F(x_k) - F(x_{k+1}) \geq \frac{3}{4}\sqrt{\frac{M}{6}}\sqrt{h_k}r_k^2 + \frac{1}{2}\lambda_k r_k^2 - \frac{\epsilon r_k}{8192},$$

which is (A.2).

If $\lambda_k = 0$, minimizing $3\sqrt{M/6}\sqrt{\epsilon}r^2/4 - \epsilon r/8192$ over $r \geq 0$ has minimizer $r = \sqrt{\epsilon}/(12288\sqrt{M/6})$ and minimum $-\epsilon^{3/2}/(201\,326\,592\sqrt{M/6})$, proving (A.3a). If $\lambda_k > 0$, then $r_k = \sqrt{h_k}/(4\sqrt{M/6})$, and

$$\frac{3}{4}\sqrt{\frac{M}{6}}\sqrt{h_k}r_k^2 - \frac{\epsilon r_k}{8192} \geq \left(\frac{3}{64} - \frac{1}{32768} \right) \frac{h_k^{3/2}}{\sqrt{M/6}} = \frac{1535}{32768} \frac{h_k^{3/2}}{\sqrt{M/6}}.$$

Together with the multiplier term in (A.2), this is (A.3b).

The common gradient estimate gives

$$\begin{aligned} \|g_{k+1}\| &\leq 2\sqrt{\frac{M}{6}}\sqrt{h_k}r_k + 2\xi_k r_k + \lambda_k r_k + \frac{M}{2}r_k^2 + \frac{\epsilon}{4096} \\ &\leq \left(\frac{1}{2} + \frac{1}{128} + \frac{3}{16} + \frac{1}{4096} \right) h_k + \lambda_k r_k \\ &= \frac{2849}{4096} h_k + \lambda_k r_k. \end{aligned}$$

Here we used, in order, $2\sqrt{M/6}\sqrt{h_k}r_k \leq h_k/2$, $2\xi_k r_k \leq h_k/128$, $(M/2)r_k^2 \leq 3h_k/16$, and $\epsilon/4096 \leq h_k/4096$. Clipping the resulting bound below by ϵ proves (A.4). $\square$

For any finite prefix of $N \leq K$ nonterminal iterations, let

$$\mathcal{B}_N := \{0 \leq k < N : \lambda_k > 0\}, \quad \mathcal{I}_N := \{0 \leq k < N : \lambda_k = 0\},$$

and, for $k \in \mathcal{B}_N$, define

$$z_k := \frac{\lambda_k r_k}{h_k}. \quad (\text{A.5})$$

Lemma A.3 (Boundary and interior budgets). *Fix $\epsilon > 0$ and suppose [Assumptions 2.1](#) and [3.1](#) hold. For the sequence generated by [\(3.3\)](#)–[\(3.4\)](#), every finite nonterminal prefix satisfies*

$$|\mathcal{B}_N| \leq \frac{32768}{1535} \sqrt{\frac{M}{6}} \Delta_F \epsilon^{-3/2} + \frac{1}{9\,431\,040} |\mathcal{I}_N|, \quad (\text{A.6a})$$

$$\sum_{k \in \mathcal{B}_N} z_k \leq 8 \sqrt{\frac{M}{6}} \Delta_F \epsilon^{-3/2} + \frac{1}{25\,165\,824} |\mathcal{I}_N|. \quad (\text{A.6b})$$

Furthermore,

$$|\mathcal{I}_N| \leq |\mathcal{B}_N| + 1 + \frac{\log_+(h_0/\epsilon) + \sum_{k \in \mathcal{B}_N} z_k}{\log(4096/2849)}. \quad (\text{A.7})$$

Proof of [Lemma A.3](#). Summing [\(A.3\)](#) and using $F(x_N) \geq F^*$ gives

$$\Delta_F \geq \frac{1535\epsilon^{3/2}}{32768\sqrt{M/6}} |\mathcal{B}_N| + \frac{1}{2} \sum_{k \in \mathcal{B}_N} \lambda_k r_k^2 - \frac{\epsilon^{3/2}}{201\,326\,592\sqrt{M/6}} |\mathcal{I}_N|.$$

This proves [\(A.6a\)](#). For $k \in \mathcal{B}_N$,

$$\lambda_k r_k^2 = \frac{z_k h_k^{3/2}}{4\sqrt{M/6}} \geq \frac{z_k \epsilon^{3/2}}{4\sqrt{M/6}},$$

which proves [\(A.6b\)](#).

For $k \in \mathcal{B}_N$, [\(A.4\)](#) implies

$$\left[\log \frac{h_{k+1}}{h_k} \right]_+ \leq \left[\log \left(\frac{2849}{4096} + z_k \right) \right]_+ \leq z_k.$$

Indeed, the left-hand side is zero when $z_k \leq 1247/4096$; above this threshold, $z - \log(2849/4096 + z)$ is increasing and positive. For $k \in \mathcal{I}_N$ with $h_{k+1} > \epsilon$, the quantity $\log h_k$ decreases by at least $\log(4096/2849)$. If $k \in \mathcal{I}_N$ has $h_{k+1} = \epsilon$ and $k+1 < N$, the gradient stopping test holds at the next iterate. Since the repeated-subproblem procedure has not terminated, $\lambda_{\min}(H_{k+1}) < -\sqrt{M}\epsilon + \xi_{k+1}$. Therefore

$$\begin{aligned} \lambda_{\min} \left(H_{k+1} + \xi_{k+1} I + 2\sqrt{\frac{M}{6}} \sqrt{\epsilon} I \right) &< -\sqrt{M}\epsilon + 2\xi_{k+1} + 2\sqrt{\frac{M}{6}} \sqrt{\epsilon} \\ &\leq \left(2 + \frac{1}{32} - \sqrt{6} \right) \sqrt{\frac{M}{6}} \sqrt{\epsilon} < 0. \end{aligned}$$

Applying [\(2.4\)](#) at iteration $k+1$ with $B = H_{k+1} + \xi_{k+1} I + 2\sqrt{M/6}\sqrt{\epsilon} I$ therefore forces $\lambda_{k+1} > 0$, so the next index belongs to $\mathcal{B}_N$. These floor-hitting indices in $\mathcal{I}_N$ pair injectively with their successors in $\mathcal{B}_N$, apart from the possible last index $k = N-1$. Hence

$$|\{k \in \mathcal{I}_N : h_{k+1} = \epsilon\}| \leq |\mathcal{B}_N| + 1.$$

For the remaining indices in $\mathcal{I}_N$, telescoping $\log h_k$, using $h_N \geq \epsilon$, and charging every positive increase on $k \in \mathcal{B}_N$ to z_k gives

$$|\{k \in \mathcal{I}_N : h_{k+1} > \epsilon\}| \log(4096/2849) \leq \log_+(h_0/\epsilon) + \sum_{k \in \mathcal{B}_N} z_k.$$

Adding the last two displays proves [\(A.7\)](#). □

Proof of Theorem 3.1. Substituting (A.6) into (A.7) gives

$$\begin{aligned} & \left[1 - \frac{1}{9\,431\,040} - \frac{1}{25\,165\,824 \log(4096/2849)} \right] |\mathcal{I}_N| \\ & \leq \left(\frac{32768}{1535} + \frac{8}{\log(4096/2849)} \right) \sqrt{\frac{M}{6}} \Delta_F \epsilon^{-3/2} + 1 + \frac{\log_+(h_0/\epsilon)}{\log(4096/2849)}. \end{aligned}$$

The elementary bounds

$$\log(4096/2849) > \frac{363}{1000}, \quad 1 - \frac{1}{9\,431\,040} - \frac{1}{25\,165\,824 \log(4096/2849)} > \frac{9\,999\,997}{10\,000\,000}$$

and (A.6a) now give, by direct substitution,

$$N = |\mathcal{I}_N| + |\mathcal{B}_N| \leq 65 \sqrt{\frac{M}{6}} \Delta_F \epsilon^{-3/2} + 3 \log_+ \frac{h_0}{\epsilon} + 2. \quad (\text{A.8})$$

This estimate is uniform over every finite nonterminal prefix. If $K = \infty$, then for every finite N the identity $N = |\mathcal{I}_N| + |\mathcal{B}_N|$ would be bounded by the same constant, independent of N . Choosing a larger N gives a contradiction. Therefore $K < \infty$. Taking $N = K$ proves (3.5) for the actual number of solved subproblems.

At termination, (3.1) gives $\|\nabla F(\hat{x})\| \leq 8193\epsilon/8192$ and, by Weyl's inequality,

$$\lambda_{\min}(\nabla^2 F(\hat{x})) \geq \lambda_{\min}(H_K) - \xi_K \geq -\sqrt{M\epsilon},$$

which proves (3.6) without a slack comparison. $\square$

Proof of Lemma 3.1. Since $\sqrt{h_k} \geq 4\sqrt{M/6} r_k$, (A.2) implies

$$F(x_k) - F(x_{k+1}) \geq \frac{M}{2} r_k^3 - \frac{\epsilon r_k}{8192}.$$

Split the first term equally and maximize $\epsilon r/8192 - M r^3/4$ over $r \geq 0$. Its maximum is $\epsilon^{3/2}/(393\,216\sqrt{6M})$, and therefore

$$F(x_k) - F(x_{k+1}) \geq \frac{M}{4} r_k^3 - \frac{\epsilon^{3/2}}{393\,216\sqrt{6M}}.$$

Summing from $k = 0$ to $K - 1$ and using $F(x_K) \geq F^*$ proves (3.7). $\square$

A.3 Certified-inexact adaptive clipped UTR

Lemma A.4 (Guaranteed trial outcome). *Suppose Assumption 4.1 holds. If*

$$\frac{M}{6} \leq \sigma^2 \leq \bar{\sigma}^2, \quad (\text{A.9})$$

then a trial of Algorithm 1 either satisfies the stopping test or satisfies the acceptance test (4.8).

Proof of Lemma A.4. If the stopping test holds, the algorithm returns before solving a subproblem. Otherwise apply Lemma A.1 with $g = g_k$, $H = H_{k,\sigma}$, $g^+ = g_k^+$, and

$$a = \sigma \sqrt{h_k} + \frac{\sigma \sqrt{\epsilon}}{64}, \quad R = \frac{\sqrt{h_k}}{4\sigma}, \quad \xi = \frac{\sigma \sqrt{\epsilon}}{64}, \quad \delta_0 = \delta_+ = \delta_g.$$

The radius and (A.9) give $(M/6)\|d\|^3 \leq \sigma \sqrt{h_k} \|d\|^2/4$. Thus (A.1b) becomes

$$F(x_k) - F(x_k + d) \geq \frac{1}{4} \sigma \sqrt{h_k} \|d\|^2 + \frac{1}{2} \lambda \|d\|^2 - \delta_g \|d\|. \quad (\text{A.10})$$

Similarly, (A.1c) yields

$$\begin{aligned}
\|g_k^+\| &\leq \sigma\sqrt{h_k}\|d\| + \lambda\|d\| + 2\frac{\sigma\sqrt{\epsilon}}{64}\|d\| + \frac{M}{2}\|d\|^2 + 2\delta_g \\
&\leq \left(\frac{1}{4} + \frac{1}{128} + \frac{3}{16} + \frac{1}{128}\right)h_k + \lambda\|d\| \\
&= \frac{29}{64}h_k + \lambda\|d\| \leq \frac{1}{2}h_k + \lambda\|d\|.
\end{aligned} \tag{A.11}$$

If $\|d\| < \sqrt{h_k}/(4\sigma)$, complementarity gives $\lambda = 0$ by Lemma 2.2. Maximizing $\delta_g r - \sigma\sqrt{h_k}r^2/4$ over $r \geq 0$ and then adding the two value-oracle errors shows

$$\tilde{F}_k^+ - \tilde{F}_k \leq \frac{\delta_g^2}{\sigma_*\sqrt{\epsilon}} + \frac{\omega}{4} \leq \omega.$$

Together with (A.11), this proves (4.8a).

If $\|d\| = \sqrt{h_k}/(4\sigma)$, then $\delta_g\|d\| \leq h_k^{3/2}/(1024\sigma)$, so (A.10) and the two value-oracle errors give

$$\tilde{F}_k - \tilde{F}_k^+ \geq \frac{15}{1024}\frac{h_k^{3/2}}{\sigma} + \frac{1}{2}\lambda\|d\|^2 - \frac{\omega}{4}.$$

Condition (4.2b), $h_k \geq \epsilon$, $\sigma \leq \bar{\sigma}$, and $\eta \leq 1/256$ imply the boundary acceptance test. The gradient test follows from (A.11). Thus every subproblem actually solved with $\sigma^2 \geq M/6$ satisfies the appropriate branch of the acceptance test, which proves the stated alternative. $\square$

Corollary A.1 (Uniform parameter bound). *Suppose Assumption 4.1 holds. Every trial parameter generated by Algorithm 1 satisfies $\sigma \leq \bar{\sigma}$, with $\bar{\sigma}$ defined in (4.1). Every inner loop terminates after finitely many trials.*

Proof of Corollary A.1. The initial parameter satisfies $\sigma_0 \leq \bar{\sigma}$. Suppose a trial parameter is at most $\bar{\sigma}$. If it is rejected, then Lemma A.4 implies $\sigma < \sigma_*$, so its successor is smaller than $\gamma_+\sigma_* \leq \bar{\sigma}$. After an accepted trial with parameter σ , the next initial value is $\max\{\sigma_{\min}, \sigma/\gamma_-\} \leq \max\{\sigma_{\min}, \sigma\}$. Since $\sigma_{\min} \leq \sigma_0$, induction proves the uniform bound. Finally, repeated rejected-trial updates reach σ_* after finitely many trials, and the first such trial lies below $\gamma_+\sigma_* \leq \bar{\sigma}$; by Lemma A.4 it cannot be rejected. Hence every inner loop is finite. $\square$

With $\inf \emptyset = \infty$, define the accepted-step return index

$$K_\epsilon := \inf\{k \in \mathbb{N}_0 : \text{Algorithm 1 returns } x_k\} \in \mathbb{N}_0 \cup \{\infty\}. \tag{A.12}$$

For every $k < K_\epsilon$, let σ_k , d_k , and λ_k denote the parameter, step, and multiplier of the accepted trial producing x_{k+1} , set $H_k := H_{k,\sigma_k}$ for its scale-dependent Hessian, and set $r_k := \|d_k\|$. For any finite $N \leq K_\epsilon$, define

$$\mathcal{B}_N := \left\{0 \leq k < N : \|d_k\| = \frac{\sqrt{h_k}}{4\sigma_k}\right\}, \quad \mathcal{I}_N := \left\{0 \leq k < N : \|d_k\| < \frac{\sqrt{h_k}}{4\sigma_k}\right\}, \tag{A.13}$$

and, for $k \in \mathcal{B}_N$, set

$$z_k := \frac{\lambda_k\|d_k\|}{h_k}. \tag{A.14}$$

Lemma A.5 (Boundary steps). *Suppose Assumption 4.1 holds. With the notation in (A.12)–(A.14), every finite $N \leq K_\epsilon$ satisfies*

$$|\mathcal{B}_N| \leq \frac{\bar{\sigma}(\Delta_F + \omega/4)}{\eta\epsilon^{3/2}} + \frac{\bar{\sigma}\omega}{\eta\epsilon^{3/2}}|\mathcal{I}_N|, \tag{A.15a}$$

$$\sum_{k \in \mathcal{B}_N} z_k \leq \frac{4\bar{\sigma}(\Delta_F + \omega/4)}{\eta\epsilon^{3/2}} + \frac{4\bar{\sigma}\omega}{\eta\epsilon^{3/2}}|\mathcal{I}_N|. \tag{A.15b}$$

Proof of Lemma A.5. For $k \in \mathcal{B}_N$, the accepted value test gives a decrease of the stored proxy of at least $\eta(h_k^{3/2}/\sigma_k + \lambda_k r_k^2) + \omega$. For $k \in \mathcal{I}_N$, that proxy can increase by at most ω . The two endpoint oracle bounds imply $\tilde{F}_0 - \tilde{F}_N \leq \Delta_F + \omega/4$. Telescoping the stored proxy therefore gives

$$\eta \sum_{k \in \mathcal{B}_N} \left(\frac{h_k^{3/2}}{\sigma_k} + \lambda_k r_k^2 \right) \leq \Delta_F + \frac{\omega}{4} + \omega |\mathcal{I}_N|.$$

Using $h_k \geq \epsilon$ and $\sigma_k \leq \bar{\sigma}$ proves (A.15a). For $k \in \mathcal{B}_N$,

$$\lambda_k \|d_k\|^2 = z_k h_k \|d_k\| = \frac{z_k h_k^{3/2}}{4\sigma_k} \geq \frac{z_k \epsilon^{3/2}}{4\bar{\sigma}}.$$

Summing the multiplier term in the same telescope proves (A.15b). $\square$

Lemma A.6 (Gradient-contraction steps). *Suppose Assumption 4.1 holds. With the notation in (A.12)–(A.14), every finite $N \leq K_\epsilon$ satisfies*

$$|\mathcal{I}_N| \leq |\mathcal{B}_N| + 1 + \frac{\log_+(h_0/\epsilon) + \sum_{k \in \mathcal{B}_N} z_k}{\log 2}. \quad (\text{A.16})$$

Proof of Lemma A.6. For $k \in \mathcal{B}_N$, (4.8c) gives

$$\frac{h_{k+1}}{h_k} \leq \max \left\{ 1, \frac{1}{2} + z_k \right\},$$

and therefore

$$\left[\log \frac{h_{k+1}}{h_k} \right]_+ \leq z_k.$$

For $k \in \mathcal{I}_N$, (4.8a) gives the clipped contraction

$$h_{k+1} = \max\{\|g_{k+1}\|, \epsilon\} \leq \max \left\{ \frac{1}{2} h_k, \epsilon \right\}.$$

Thus an index $k \in \mathcal{I}_N$ with $h_{k+1} > \epsilon$ satisfies $\log(h_{k+1}/h_k) \leq -\log 2$. The complementary case $h_{k+1} = \epsilon$ means precisely that the clipping floor is active, or equivalently that $\|g_{k+1}\| \leq \epsilon$; the threshold is ϵ , rather than $\epsilon/2$, because h_{k+1} is clipped below at ϵ .

Suppose now that $k \in \mathcal{I}_N$ has $h_{k+1} = \epsilon$ and $k+1 < N$. Then the algorithm does not return at x_{k+1} and the next outer iteration accepts a step. In particular, the stopping test fails at its accepted parameter σ_{k+1} . Since $\|g_{k+1}\| \leq \epsilon$, this failure gives

$$\lambda_{\min} \left(H_{k+1} + \frac{65}{64} \sigma_{k+1} \sqrt{\epsilon} I \right) < 0.$$

Indeed, failure of the Hessian stopping test gives $\lambda_{\min}(H_{k+1}) < -(9/8)\sigma_{k+1}\sqrt{\epsilon}$, and $65/64 < 9/8$. Since $h_{k+1} = \epsilon$, applying the positive-semidefiniteness condition in (2.4) with $B = H_{k+1} + (65/64)\sigma_{k+1}\sqrt{\epsilon}I$ forces $\lambda_{k+1} > 0$. Complementarity then gives $k+1 \in \mathcal{B}_N$. Pairing each such floor-hitting index with its successor is injective, and only $k = N-1$ can remain unpaired. Therefore

$$|\{k \in \mathcal{I}_N : h_{k+1} = \epsilon\}| \leq |\mathcal{B}_N| + 1.$$

For the same floor-hitting steps, $\log(h_{k+1}/h_k) \leq 0$. Telescoping $\log h_k$ from 0 to N and using $h_N \geq \epsilon$ now gives

$$|\{k \in \mathcal{I}_N : h_{k+1} > \epsilon\}| \log 2 \leq \log \frac{h_0}{\epsilon} + \sum_{k \in \mathcal{B}_N} z_k = \log_+(h_0/\epsilon) + \sum_{k \in \mathcal{B}_N} z_k.$$

Adding the two disjoint interior-step counts proves (A.16). $\square$

Combining [Lemmas A.5](#) and [A.6](#) with (4.3) gives, for every finite prefix of N accepted nonterminal steps,

$$|\mathcal{I}_N| \leq \frac{(1 + 4/\log 2)\bar{\sigma}(\Delta_F + \omega/4)/(\eta\epsilon^{3/2}) + 1 + \log_+(h_0/\epsilon)/\log 2}{1 - \chi}, \quad (\text{A.17a})$$

$$N \leq \frac{\bar{\sigma}(\Delta_F + \omega/4)}{\eta\epsilon^{3/2}} + \left(1 + \frac{\bar{\sigma}\omega}{\eta\epsilon^{3/2}}\right) |\mathcal{I}_N|. \quad (\text{A.17b})$$

Lemma A.7 (Rejected trials). *Suppose [Assumption 4.1](#) holds. Let K_ϵ be the finite return index in (A.12), and let U_ϵ be the number of rejected trials before termination. Then*

$$U_\epsilon \leq \frac{\log \gamma_-}{\log \gamma_+} K_\epsilon + \frac{\log(\bar{\sigma}/\sigma_0)}{\log \gamma_+}. \quad (\text{A.18})$$

Proof of [Lemma A.7](#). For $0 \leq k \leq K_\epsilon$, let σ_k^{init} be the initial parameter of outer iteration k , and let u_k be its number of rejected trials. For $0 \leq k < K_\epsilon$, the accepted parameter is $\sigma_k = \sigma_k^{\text{init}} \gamma_+^{u_k}$, so the update rule gives

$$\sigma_{k+1}^{\text{init}} \geq \frac{\sigma_k^{\text{init}} \gamma_+^{u_k}}{\gamma_-},$$

and hence

$$u_k \log \gamma_+ \leq \log \sigma_{k+1}^{\text{init}} - \log \sigma_k^{\text{init}} + \log \gamma_-.$$

Summing over $0 \leq k < K_\epsilon$ and using $\sigma_0^{\text{init}} = \sigma_0$ yields

$$\sum_{k=0}^{K_\epsilon-1} u_k \log \gamma_+ \leq \log \frac{\sigma_{K_\epsilon}^{\text{init}}}{\sigma_0} + K_\epsilon \log \gamma_-. \quad (\text{A.19})$$

In the terminal inner loop, the stopping test returns at parameter $\sigma_{K_\epsilon}^{\text{init}} \gamma_+^{u_{K_\epsilon}} \leq \bar{\sigma}$ by [Corollary A.1](#). Thus

$$u_{K_\epsilon} \log \gamma_+ \leq \log \frac{\bar{\sigma}}{\sigma_{K_\epsilon}^{\text{init}}}.$$

Since $U_\epsilon = \sum_{k=0}^{K_\epsilon-1} u_k + u_{K_\epsilon}$, adding this inequality to (A.19) proves (A.18). $\square$

Proof of [Theorem 4.1](#). Combining [Lemmas A.5](#) and [A.6](#) and using (4.3) gives (A.17) for every finite nonterminal prefix. The right-hand side is independent of the prefix length. Hence $K_\epsilon = \infty$ would give the same finite upper bound on $N = |\mathcal{B}_N| + |\mathcal{I}_N|$ for arbitrarily large N , which is impossible. Thus $K_\epsilon < \infty$, and taking $N = K_\epsilon$ establishes the displayed accepted-step bounds.

The final stopping test solves no subproblem. Therefore $Q_\epsilon = K_\epsilon + U_\epsilon$, and [Lemma A.7](#) gives the trial bound. At termination, the current parameter is at most $\bar{\sigma}$ by [Corollary A.1](#). The gradient oracle error gives the first inequality in (4.10). Weyl's inequality, the Hessian stopping test, and (4.5) give

$$\lambda_{\min}(\nabla^2 F(\hat{x})) \geq -\left(\frac{9}{8} + \frac{1}{64}\right) \bar{\sigma} \sqrt{\epsilon} = -\frac{73}{64} \bar{\sigma} \sqrt{\epsilon}.$$

$\square$

A.4 NC–SC minimax results

Proof of [Lemma 5.2](#). Strong concavity and Cauchy–Schwarz give the first inequality. The second follows from ℓ -smoothness and $\nabla_y f(x, y^*(x)) = 0$. $\square$

Proof of Proposition 5.1. Apply the standard AGD estimate to the smooth strongly convex objective

$$\phi_x(y) := -f(x, y).$$

See Wang and Li [10, Lemma 2] or Luo et al. [4, Lemma 5]. By (5.5), the initial distance is at most R_{in}/μ . It is sufficient to reduce the distance to τ/ℓ , because ℓ -smoothness then gives the requested residual. Combining these facts with $1 - 1/\sqrt{\kappa} \leq \exp(-1/\sqrt{\kappa})$ proves (5.10). $\square$

Proof of Lemma 5.3. Corrected value. Fix x , write $C = \nabla_{yy}^2 f(x, y)$, $q = \nabla_y f(x, y)$, and $\delta = y - y^*(x)$. Taylor expansion of the inner gradient gives

$$v := q - C\delta, \quad \|v\| \leq \frac{\rho}{2} \|\delta\|^2.$$

If

$$\overline{C} := \int_0^1 \nabla_{yy}^2 f(x, y - t\delta) dt,$$

then the fundamental theorem of calculus gives $q = \overline{C}\delta$ and hence $v = (\overline{C} - C)\delta$. Both matrices satisfy $-\ell I \preceq C \preceq -\mu I$ and $-\ell I \preceq \overline{C} \preceq -\mu I$, so

$$\|v\| \leq (\ell - \mu) \|\delta\|.$$

Taylor expansion of the payoff gives

$$P(x) = f(x, y) - q^T \delta + \frac{1}{2} \delta^T C \delta + R, \quad |R| \leq \frac{\rho}{6} \|\delta\|^3.$$

Using $q = C\delta + v$ and completing the square yields

$$P(x) - \hat{P}(x, y) = R + \frac{1}{2} v^T C^{-1} v.$$

Strong concavity gives $\|C^{-1}\| \leq 1/\mu$. Moreover,

$$\|v\|^2 \leq \frac{\rho}{2} \|\delta\|^2 (\ell - \mu) \|\delta\|,$$

so

$$\frac{1}{2} |v^T C^{-1} v| \leq \frac{(\kappa - 1)\rho}{4} \|\delta\|^3.$$

Together with the Taylor remainder, this proves (5.13a); its second inequality follows from $\kappa \geq 1$. If $\|\delta\| \leq \mu/\rho$, then the quadratic Taylor bound on v gives

$$\frac{1}{2} |v^T C^{-1} v| \leq \frac{\rho}{8} \|\delta\|^3.$$

Combining this estimate with the bound on R proves the κ -independent local estimate (5.14).

Corrected gradient. Let

$$y_t := y^*(x) + t(y - y^*(x)), \quad t \in [0, 1].$$

Since f is twice continuously differentiable, the fundamental theorem of calculus applied to the two partial gradients along this segment gives the exact identities

$$\begin{aligned} \nabla_x f(x, y) - \nabla_x f(x, y^*(x)) &= \int_0^1 \nabla_{xy}^2 f(x, y_t) (y - y^*(x)) dt, \\ \nabla_y f(x, y) &= \int_0^1 \nabla_{yy}^2 f(x, y_t) (y - y^*(x)) dt, \end{aligned}$$

where the second identity uses $\nabla_y f(x, y^*(x)) = 0$. Substituting these identities into (5.12a) and using (5.3a), we obtain

$$\begin{aligned}\widehat{g}(x, y) - \nabla P(x) &= \int_0^1 \left[\nabla_{xy}^2 f(x, y_t) - \nabla_{xy}^2 f(x, y) \right. \\ &\quad \left. + \nabla_{xy}^2 f(x, y) (\nabla_{yy}^2 f(x, y))^{-1} (\nabla_{yy}^2 f(x, y) - \nabla_{yy}^2 f(x, y_t)) \right] \\ &\quad \times (y - y^*(x)) dt.\end{aligned}$$

This identity displays the cancellation of the linear inner error. Hessian Lipschitzness, together with

$$\left\| \nabla_{xy}^2 f(x, y) (\nabla_{yy}^2 f(x, y))^{-1} \right\| \leq \kappa,$$

bounds the norm of the integrand by

$$(1 + \kappa)\rho(1 - t)\|y - y^*(x)\|^2.$$

Integrating over $t \in [0, 1]$ proves (5.13b).

Corrected Hessian. Retain $C = \nabla_{yy}^2 f(x, y)$ and set

$$B := \nabla_{xy}^2 f(x, y), \quad B_* := \nabla_{xy}^2 f(x, y^*(x)), \quad C_* := \nabla_{yy}^2 f(x, y^*(x)).$$

The inverse identity gives

$$\begin{aligned}\left\| (\nabla_{yy}^2 f(x, y))^{-1} - (\nabla_{yy}^2 f(x, y^*(x)))^{-1} \right\| \\ \leq \frac{\rho}{\mu^2} \|y - y^*(x)\|.\end{aligned}$$

The Schur-complement products satisfy the exact telescoping identity

$$\begin{aligned}BC^{-1}B^T - B_*C_*^{-1}B_*^T &= (B - B_*)C^{-1}B^T + B_*(C^{-1} - C_*^{-1})B^T \\ &\quad + B_*C_*^{-1}(B - B_*)^T.\end{aligned}$$

Hessian Lipschitzness, $\|B\|, \|B_*\| \leq \ell$, and $\|C^{-1}\|, \|C_*^{-1}\| \leq 1/\mu$ bound these three terms by $\kappa\rho\|y - y^*(x)\|$, $\kappa^2\rho\|y - y^*(x)\|$, and $\kappa\rho\|y - y^*(x)\|$, respectively. Adding the xx -block error gives

$$\begin{aligned}\|\widehat{H}(x, y) - \nabla^2 P(x)\| &\leq \rho\|y - y^*(x)\| + 2\frac{\ell}{\mu}\rho\|y - y^*(x)\| \\ &\quad + \frac{\ell^2}{\mu^2}\rho\|y - y^*(x)\| \\ &= (1 + \kappa)^2\rho\|y - y^*(x)\|,\end{aligned}$$

which proves (5.13c). □

Proof of Corollary 5.1. The residual-distance certificate gives

$$\|y - y^*(x)\| \leq \frac{\|\nabla_y f(x, y)\|}{\mu}.$$

Consequently, (5.13c) gives $\|\widehat{H} - \nabla^2 P(x)\| \leq \xi(x, y)$. Moreover, (5.15) gives

$$\|y - y^*(x)\| \leq \frac{\sqrt{M_P\epsilon/6}}{64(1 + \kappa)^2\rho}.$$

Substitution in (5.13c) gives the upper bound on $\xi(x, y)$ in (5.17). Substitution in (5.13b) gives

$$\|\widehat{g} - \nabla P\| \leq \frac{M_P\epsilon}{49152(1 + \kappa)^3\rho} = \frac{\sqrt{2}}{12288} \left(\frac{\kappa}{1 + \kappa} \right)^3 \epsilon < \frac{\epsilon}{8192}.$$

These bounds are stronger than (3.1). □

Proof of Corollary 5.3. Equation (5.5) gives $\|y - y^*(x)\| \leq \delta_{VG}$ or $\delta_H(\sigma)$, respectively. The first branch of (5.24a) and (5.13b) give the gradient certificate. The second branch and (5.13a) make the cubic value error at most $\omega/8$. Equation (5.24b) and (5.13c) give the Hessian certificate. $\square$

Verification of (5.27) and (5.28). For (5.26), the interior condition holds with left-hand side at most $3\omega/4$, while

$$\chi = \frac{1}{16} \left(1 + \frac{4}{\log 2} \right) < 1$$

and the boundary condition follows from $15/1024 - 1/256 = 11/1024 > 5/16384$. Finally, Lemma 5.1 gives the required regularity of P and Assumption 5.1 gives $P^* > -\infty$. Applying Theorem 4.1, and using $\|\hat{g}_0\| \leq \|\nabla P(x_0)\| + \epsilon/256$ to simplify its initial logarithm, proves (5.27) and (5.28). $\square$

Proof of Theorem 5.1. Let K be the number of nonterminal outer iterations and write $r_k := \|d_k\|$. The residual contract and Corollary 5.1 give

$$\|g_k - \nabla P(x_k)\| \leq \epsilon/8192, \quad \|H_k - \nabla^2 P(x_k)\| \leq \xi_k \leq \frac{1}{64} \sqrt{\frac{M_P}{6}} \sqrt{\epsilon}.$$

Thus the outer sequence obeys the fixed-scale repeated-subproblem procedure in Section 3, and Theorem 3.1 applies with $M = M_P$. In particular, (3.5) gives (5.21). Its logarithmic term uses $h_0 = \max\{\|g_0\|, \epsilon\}$. Since

$$\|g_0\| \leq \|\nabla P(x_0)\| + \epsilon/8192,$$

we have $\log_+(h_0/\epsilon) = O(\log_+(\|\nabla P(x_0)\|/\epsilon) + 1)$. Together with $\sqrt{M_P} = \sqrt{4\sqrt{2}} \kappa^{3/2} \sqrt{\rho}$, this proves (5.19) and (5.20).

It remains to transfer the work contract of $\mathcal{S}$. At the start of the inner call for x_{k+1} , joint ℓ -smoothness and the previous residual guarantee give the directly observable warm-start bound

$$\|\nabla_y f(x_{k+1}, y_k)\| \leq \tau_{\text{out}} + \ell r_k.$$

Consequently, apart from the initial call, the logarithmic term in (5.7) is bounded by

$$1 + \log_2 \left(1 + \frac{\ell r_k}{\tau_{\text{out}}} \right).$$

By (5.18b),

$$\frac{\ell r_k}{\tau_{\text{out}}} \leq 64\sqrt{3} r_k \sqrt{\frac{M_P}{\epsilon}}.$$

Using $\log_2(1+t) = O(1+t^3)$ for $t \geq 0$, the logarithm in each noninitial call is therefore

$$O \left(1 + \left(r_k \sqrt{\frac{M_P}{\epsilon}} \right)^3 \right).$$

By Lemma 3.1,

$$\begin{aligned} \sum_{k=0}^{K-1} \left(r_k \sqrt{\frac{M_P}{\epsilon}} \right)^3 &= \frac{M_P^{3/2}}{\epsilon^{3/2}} \sum_{k=0}^{K-1} r_k^3 \\ &\leq 4\Delta_P \sqrt{M_P} \epsilon^{-3/2} + \frac{K}{98304\sqrt{6}}. \end{aligned}$$

Summing the module work bounds, adding the initial call, and substituting the outer bound for K proves (5.22). Finally, $\sqrt{M_P} = \sqrt{4\sqrt{2}} \kappa^{3/2} \sqrt{\rho}$ proves (5.23). $\square$

A.5 Parameter-free residual-module UTR

Source and proof sketch for Theorem 6.1. This is Theorem 5.1 of Lan et al. [9]. For completeness, its counting argument is as follows. An accelerated-restart call at the current strong-convexity guess μ_t either detects failure and replaces the guess by $\mu_t/4$, or reduces the gradient norm by a factor two. Once $\mu_t \leq \mu$, strong convexity converts the distance estimate produced by the accelerated subroutine into the latter gradient reduction. Thus there are at most $\lceil \log_4(\mu_0/\mu) \rceil$ failed guesses and at most $\lceil \log_2(\|\nabla\phi(u_0)\|/\tau) \rceil$ successful reductions. The accepted guesses remain at least $\mu/4$, so summing the $O(\sqrt{L/\mu_t})$ cost of the accelerated calls gives (6.2). At the endpoint $L = \mu$, strong monotonicity, Lipschitz continuity, and Cauchy–Schwarz force $\nabla\phi(u) - \nabla\phi(v) = L(u - v)$ for every u, v . Taking v to be the minimizer proves the one-step claim. $\square$

Proof of Corollary 6.1. Strong monotonicity and Lipschitz continuity of ∇q_x imply

$$\mu\|z - y^{\text{in}}\| \leq \|\nabla q_x(z) - \nabla q_x(y^{\text{in}})\| \leq \ell\|z - y^{\text{in}}\|.$$

Consequently, $\mu \leq a_x \leq \ell$. The initialization satisfies the hypotheses of Theorem 6.1 when $\kappa > 1$, and $\log_4(a_x/\mu) \leq \log_4 \kappa$. Substitution in (6.2) proves the claim in that case. If $\kappa = 1$, then $a_x = \mu = \ell$ and the endpoint argument following Theorem 6.1 gives a numerical constant per call. $\square$

Proof of Lemma 6.1. Set $\kappa := \ell/\mu$ and $s := \sqrt{M_P/6}$. Choose ϵ_{or} to be

$$\min \left\{ 1, \frac{\mu}{\sqrt{\rho}}, \left(\frac{\mu^2}{128(1+\kappa)\rho} \right)^{1/3}, \left(\frac{s\mu}{64(1+\kappa)^2\rho} \right)^{2/3}, \left(\frac{3\mu^3}{7\rho} \right)^{1/3} \right\}. \quad (\text{A.20})$$

The residual–distance inequality in (5.5) gives $\|y - y^*(x)\| \leq \epsilon^2/\mu$. Hence (A.20) ensures $\epsilon^2/\mu \leq \mu/\rho$, and Lemma 5.3 gives

$$\begin{aligned} \|\hat{g} - \nabla P\| &\leq \frac{(1+\kappa)\rho}{2\mu^2} \epsilon^4, \\ \|\hat{H} - \nabla^2 P\| &\leq \frac{(1+\kappa)^2\rho}{\mu} \epsilon^2, \\ |\hat{P} - P| &\leq \frac{7\rho}{24\mu^3} \epsilon^6. \end{aligned}$$

The remaining three conditions in (A.20) bound these quantities by $\epsilon/256$, $s\sqrt{\epsilon}/64$, and $\epsilon^3/8$, respectively. Every coefficient depends only on (ℓ, μ, ρ) , so the choice is uniform over the entire parameter class and over all x and y . $\square$

Proof of Lemma 6.2. Apply Lemma A.1 with $F = P$, $M = M_P$,

$$a = \sigma\sqrt{h} + \frac{\sigma\sqrt{\epsilon}}{64}, \quad R = \frac{\sqrt{h}}{4\sigma}, \quad \xi = \frac{1}{64}\sqrt{\frac{M_P\epsilon}{6}}, \quad \delta_0 = \delta_+ = \frac{(1+\kappa)\rho}{2\mu^2} \epsilon^4.$$

Lemma 5.1 supplies the required Hessian regularity. At a safe scale, $\xi \leq \sigma\sqrt{\epsilon}/64$, so $a - \xi \geq \sigma\sqrt{h}$ and $a + \xi \leq \sigma\sqrt{h} + 2\sigma\sqrt{\epsilon}/64$. The radius and (6.7) imply $(M_P/6)r^3 \leq \sigma\sqrt{h}r^2/4$. Thus (A.1b) proves (6.8a).

If the radius is active, then $\sigma\sqrt{h}r^2 = h^{3/2}/(16\sigma)$. By (6.5a), the last term of (6.8a) is at most $h^{3/2}/(1024\sigma)$. This proves (6.9).

The common candidate-gradient estimate gives

$$\|g^+\| \leq \sigma\sqrt{h}r + \lambda r + 2\frac{\sigma\sqrt{\epsilon}}{64}r + \frac{(1+\kappa)\rho}{\mu^2} \epsilon^4 + \frac{M_P}{2}r^2.$$

The radius, safe-scale condition, and (6.5a) bound the four terms other than λr by, respectively, $h/4$, $h/128$, $h/128$, and $3h/16$. Their sum is $116h/256 < 117h/256$, proving (6.8b). $\square$

Proof of Lemma 6.3. Denote the current corrected gradient, accepted step, trial scale, and multiplier of accepted step k by g_k, d_k, σ_k , and λ_k , and set $h_k := \max\{\|g_k\|, \epsilon\}$ and $r_k := \|d_k\|$. For the stated prefix, define

$$\mathcal{B}_N := \left\{0 \leq k < N : r_k = \frac{\sqrt{h_k}}{4\sigma_k}\right\}, \quad \mathcal{I}_N := \left\{0 \leq k < N : r_k < \frac{\sqrt{h_k}}{4\sigma_k}\right\}.$$

Every $k \in \mathcal{B}_N$ gives a true decrease of at least

$$\eta \left(\frac{h_k^{3/2}}{\sigma_k} + \lambda_k r_k^2 \right),$$

because the two corrected-value errors sum to at most $\epsilon^3/4$. Every $k \in \mathcal{I}_N$ can increase the true value by at most $5\epsilon^3/4$. Telescoping and using $h_k \geq \epsilon$, $\sigma_k \leq \bar{\sigma}$ give

$$|\mathcal{B}_N| \leq \frac{\bar{\sigma}\Delta_P}{\eta\epsilon^{3/2}} + \frac{5\bar{\sigma}\epsilon^{3/2}}{4\eta} |\mathcal{I}_N|, \quad (\text{A.21a})$$

$$\sum_{k \in \mathcal{B}_N} \frac{\lambda_k r_k}{h_k} \leq \frac{4\bar{\sigma}\Delta_P}{\eta\epsilon^{3/2}} + \frac{5\bar{\sigma}\epsilon^{3/2}}{\eta} |\mathcal{I}_N|. \quad (\text{A.21b})$$

Here we used, for $k \in \mathcal{B}_N$, $\lambda_k r_k^2 = (\lambda_k r_k / h_k) h_k^{3/2} / (4\sigma_k)$.

The accepted gradient tests give

$$h_{k+1} \leq \begin{cases} \max\{h_k/2, \epsilon\}, & k \in \mathcal{I}_N, \\ \max\{h_k/2 + \lambda_k r_k, \epsilon\}, & k \in \mathcal{B}_N. \end{cases}$$

Thus every $k \in \mathcal{I}_N$ not hitting the floor ϵ decreases $\log(h_k/\epsilon)$ by at least $\log 2$, whereas $k \in \mathcal{B}_N$ can increase it by at most $\lambda_k r_k / h_k$.

It remains to count floor-hitting indices in $\mathcal{I}_N$. If such an index satisfies $k+1 < N$, then at the next accepted outer iteration $h = \epsilon$ and the stopping test fails. The gradient branch of that test already holds; otherwise $h > \epsilon$, contrary to the floor event. Hence the Hessian branch must fail. To show that the next accepted solution is on the boundary, let

$$A = H + \frac{\sigma\sqrt{\epsilon}}{64}I + \sigma\sqrt{\epsilon}I.$$

Failure of the stopping test gives $-\lambda_{\min}(A) > 7\sigma\sqrt{\epsilon}/64$. Applying (2.4) with $B = A$ therefore forces $\lambda > 0$, and complementarity forces $r = \sqrt{\epsilon}/(4\sigma)$. Rejected trials only increase σ and retest stopping, so each nonterminal floor event is injectively followed by an index in $\mathcal{B}_N$; only the last prefix index can remain unpaired. Therefore

$$|\mathcal{I}_N| \leq |\mathcal{B}_N| + 1 + \frac{\log_+(\|g_0\|/\epsilon) + \sum_{k \in \mathcal{B}_N} \lambda_k r_k / h_k}{\log 2}. \quad (\text{A.22})$$

Substituting (A.21) into (A.22) and applying (6.10) bounds $N = |\mathcal{B}_N| + |\mathcal{I}_N|$ by the right-hand side of (6.11), uniformly in N . $\square$

Proof of Theorem 6.2. Set

$$s := \sqrt{M_P/6}, \quad \bar{\sigma} := 2s.$$

Choose the class-uniform threshold no larger than ϵ_{or} in Lemma 6.1 and so that

$$\begin{aligned} \epsilon &\leq s, & \frac{5\bar{\sigma}\epsilon^3}{4\eta\epsilon^{3/2}} \left(1 + \frac{4}{\log 2}\right) &\leq \frac{1}{2}, \\ \frac{1}{s\sqrt{\epsilon}} \left(\frac{(1+\kappa)\rho}{2\mu^2}\epsilon^4\right)^2 &\leq \frac{3}{4}\epsilon^3, \\ \left(\frac{1}{128} - \eta\right) \frac{\epsilon^{3/2}}{\bar{\sigma}} &\geq \frac{5}{4}\epsilon^3. \end{aligned} \quad (\text{A.23})$$

For example, take

$$\epsilon_0 := \frac{1}{2} \min \left\{ \epsilon_{\text{or}}, s, \left(\frac{\eta}{5s(1+4/\log 2)} \right)^{2/3}, \left(\frac{3s\mu^4}{(1+\kappa)^2\rho^2} \right)^{2/9} \right\}. \quad (\text{A.24})$$

For every $0 < \epsilon \leq \epsilon_0$, the term containing $\eta/[5s(1+4/\log 2)]$ implies both the absorption and boundary-acceptance conditions in (A.23); the last term implies its interior-increase condition. Thus this single choice, which depends only on (ℓ, μ, ρ) , ensures all the displayed requirements.

Starting from $\sigma = \epsilon$, a rejected trial doubles σ . Unless the algorithm has already returned at its stopping or zero-step test, the first nonzero trial with $\sigma \geq s$ has $\sigma \leq \bar{\sigma}$. By Lemma 6.2, its corrected-gradient test passes. An interior trial can have a small true increase because of the remaining gradient error. Minimizing the right-hand side of (6.8a) over $r \geq 0$ shows that this increase is at most

$$\frac{1}{\sigma\sqrt{h}} \left(\frac{(1+\kappa)\rho}{2\mu^2} \epsilon^4 \right)^2 = O(\epsilon^{15/2}).$$

Here we used $h \geq \epsilon$ and $\sigma \geq s$. This is at most $3\epsilon^3/4$ by (A.23); after adding the two value errors, the interior value test passes. On a boundary trial, (6.9) and the two value errors imply

$$\hat{P} - \hat{P}^+ \geq \frac{h^{3/2}}{128\sigma} + \frac{1}{4}\lambda r^2 - \frac{\epsilon^3}{4}.$$

Because $h \geq \epsilon$ and $\sigma \leq \bar{\sigma}$, the right-hand side is at least $\eta(h^{3/2}/\sigma + \lambda r^2) + \epsilon^3$ by (A.23). Hence a safe-scale trial either stops or is accepted, and no generated scale exceeds $\bar{\sigma}$.

The absorption condition in Lemma 6.3 follows from (A.23). Consider an arbitrary finite trial prefix, and let K_N and U_N be its numbers of accepted and rejected nonterminal trials. Applying Lemma 6.3 to its accepted-step subsequence bounds K_N by the right-hand side of (6.11), independently of the prefix length. Starting from $\sigma = \epsilon$, each rejection doubles σ and each acceptance decreases it by at most a factor of two. Since every generated scale is at most $\bar{\sigma}$, this gives

$$U_N \leq K_N + \log_2(\bar{\sigma}/\epsilon) + 1.$$

Thus the length $K_N + U_N$ of every finite trial prefix has one common finite upper bound. An infinite run would have arbitrarily long finite prefixes, a contradiction. The algorithm therefore terminates. Applying the same bounds to the complete finite run, with K accepted and U rejected trials, proves the same estimate with $\log_+(\|g_0\|/\epsilon)$ in place of the first logarithm in (6.13), because $Q \leq K + U + 1$; the extra unit accounts for a terminal solved subproblem with $d = 0$. At the initial tuple, (6.5a) gives $\|g_0\| \leq \|\nabla P(x_0)\| + \epsilon/256$. Hence replacing $\log_+(\|g_0\|/\epsilon)$ by $\log_+(\|\nabla P(x_0)\|/\epsilon)$ costs at most a universal additive constant, which proves (6.13).

If the explicit stopping test returns, then (6.5a), $\sigma \leq 2s$, and (6.5b) give

$$\|\nabla P(x)\| \leq \frac{257}{256}\epsilon, \quad \lambda_{\min}(\nabla^2 P(x)) \geq -\frac{145}{64}s\sqrt{\epsilon} \geq -\sqrt{M_P\epsilon},$$

where $145 < 64\sqrt{6}$. If the robust subproblem returns $d = 0$, global optimality of the differentiable quadratic model gives $g = 0$, $h = \epsilon$, and

$$H + \frac{\sigma\sqrt{\epsilon}}{64}I + \sigma\sqrt{\epsilon}I \succeq 0.$$

Together with (6.5) and $\sigma \leq 2s$, this gives

$$\|\nabla P(x)\| \leq \frac{\epsilon}{256}, \quad \lambda_{\min}(\nabla^2 P(x)) \geq -\frac{131}{64}s\sqrt{\epsilon} \geq -\sqrt{M_P\epsilon},$$

where $131 < 64\sqrt{6}$. A constant rescaling of the internal tolerance converts the first inequality in (6.12) to $\|\nabla P(\hat{x})\| \leq \epsilon$.

Finally, $\sqrt{M_P} = \sqrt{4\sqrt{2}}\kappa^{3/2}\sqrt{\rho}$ gives the stated leading outer dependence. $\square$

Proof of Corollary 6.2. Every accepted boundary step decreases the true value, while every accepted interior step can increase it by at most $5\epsilon^3/4$. There are at most Q accepted steps, so every accepted outer point satisfies

$$P(x) - P^* \leq \Delta_P + \frac{5}{4}Q\epsilon^3.$$

Since ∇P is $(1 + \kappa)\ell$ -Lipschitz, the standard descent inequality and (6.5a) give $h \leq G_Q$.

At a candidate point, the SCAR module is warm-started from an inner point with residual at most ϵ^2 . The joint ℓ -smoothness and the trust-region radius give the initial candidate residual bound

$$\epsilon^2 + \ell\|d\| \leq \epsilon^2 + \frac{\ell\sqrt{G_Q}}{4\epsilon}.$$

Apply Corollary 6.1 with target ϵ^2 to every candidate call and separately to the initial call at (x_0, y_{-1}) . There is at most one candidate call per solved subproblem, so summation gives (6.15). $\square$

A.6 Adaptive fixed-radius SCAR trust region

Proof of Lemma 7.1. The AR estimate used in the proof of Lan et al. [9, Theorem 5.1], which follows from their Theorem 4.1 and remains valid when $L = \mu$, states that a call with the current guess ν and an admissible line-search estimate uses at most

$$4 + C_1\sqrt{10L/\nu}$$

gradient evaluations and returns a point v satisfying

$$\|\nabla\phi_j(v)\| \leq \frac{\nu}{2}\|u - u_j^*\|, \quad u_j^* := \arg \min_u \phi_j(u).$$

If $\nu \leq \mu$, strong convexity gives $\|u - u_j^*\| \leq \|\nabla\phi_j(u)\|/\mu$, so the returned residual is at most $R/2$. Therefore a failed halving test implies $\nu > \mu$. Every failure divides ν by four, and once $\nu \leq \mu$ no later function in the sequence can cause another failure. This proves (7.1). Moreover, throughout the process $\nu \geq \mu/4$. Hence each attempted stage costs at most

$$4 + C_1\sqrt{40\kappa} \leq (4 + 2\sqrt{10}C_1)\sqrt{\kappa}.$$

There are N successful attempts and at most the number of failed attempts in (7.1), which proves (7.2). The line-search statement is the uniform estimate in Theorem 6.1. Since all ϕ_j have the same global smoothness bound L , a retained estimate remains an admissible initialization after the objective changes. $\square$

Proof of Lemma 7.2. Lipschitz continuity of the full gradient gives

$$\|\nabla_y f(x + d, y)\| \leq \|\nabla_y f(x, y)\| + \ell\|d\| \leq (1 + c_0)\ell R(\sigma).$$

After N_w successful halvings this is at most $c_0\ell R(\sigma)$ by (7.4). At the same x , one halving reduces the old residual bound to $c_0\ell R(\sigma)/2 = c_0\ell R(2\sigma)$. $\square$

Proof of Lemma 7.3. Because $c_0\kappa\sqrt{\epsilon}\mathcal{A}_\epsilon \rightarrow 0$ as $\epsilon \downarrow 0$, there is a threshold depending only on (ℓ, μ, ρ) below which (7.6) and $\sigma \geq \mathcal{A}_\epsilon^{-1}$ imply $\|y - y^*(x)\| \leq \mu/\rho$. Substituting (7.6) into (5.14) then gives (7.7), proving part (i).

For part (ii), Lemma 5.3 and (7.6) imply

$$\begin{aligned} \|\hat{g} - \nabla P\| &\leq \frac{(1 + \kappa)\rho}{2} \frac{c_0^2\kappa^2\epsilon}{\sigma^2}, \\ \|\hat{H} - \nabla^2 P\| &\leq (1 + \kappa)^2\rho \frac{c_0\kappa\sqrt{\epsilon}}{\sigma}. \end{aligned}$$

Using $\kappa \geq 1$, $M_P = 4\sqrt{2}\kappa^3\rho$, and $M_P \leq 6\sigma^2$, the right-hand sides are at most

$$\frac{3c_0^2}{2\sqrt{2}}\epsilon, \quad 3\sqrt{2}c_0\sigma\sqrt{\epsilon},$$

respectively. With $c_0 = 2^{-12}$ these are bounded by the first two quantities in (7.9). For the value bound, the ratio of (7.7) to $\epsilon^{3/2}/\sigma$ is at most $7c_0^3/(1024\sqrt{2})$, which is smaller than $1/4096$. This proves the third inequality.

For part (iii), $\mu \leq a_x \leq \ell$ and (5.5) give

$$\|y_v - y^*(x)\| \leq c_0\kappa\epsilon^{3/2}.$$

The corrected-gradient and corrected-Hessian errors are, uniformly in x , at most

$$\frac{(1+\kappa)\rho c_0^2\kappa^2}{2}\epsilon^3, \quad (1+\kappa)^2\rho c_0\kappa\epsilon^{3/2},$$

respectively. The first is at most $\epsilon/1024$ for sufficiently small ϵ . For the second, note that $\epsilon\mathcal{A}_\epsilon \rightarrow 0$ and $\sigma \geq \mathcal{A}_\epsilon^{-1}$, so the displayed Hessian error is at most $\sigma\sqrt{\epsilon}/32$ for all sufficiently small ϵ . This proves (7.11). All estimates used only the common constants (ℓ, μ, ρ) , so the thresholds are uniform over the stated class. $\square$

Proof of Lemma 7.4. Take $\epsilon_3 \leq \min\{\epsilon_1, \epsilon_2\}$ from Lemma 7.3. Let

$$m(d) := \hat{g}^T d + \frac{1}{2}d^T \left(\hat{H} + \frac{65}{64}\sigma\sqrt{\epsilon}I \right) d.$$

Apply Lemma A.1 with $F = P$, $M = M_P$,

$$a = \frac{65}{64}\sigma\sqrt{\epsilon}, \quad R = R(\sigma), \quad \xi = \frac{\sigma\sqrt{\epsilon}}{64}, \quad \delta_0 = \frac{\epsilon}{1024}, \quad \delta_+ = 0,$$

and $g^+ = \nabla P(x+d)$. The model part of the common estimate gives

$$m(d) \leq -\frac{\lambda}{2}\|d\|^2. \tag{A.25}$$

At a safe scale, (7.9), the Taylor bound (2.3b), and (A.25) yield, with $r = \|d\|$,

$$P(x+d) - P(x) \leq -\frac{\lambda}{2}r^2 - \frac{65}{128}\sigma\sqrt{\epsilon}r^2 + \frac{\epsilon}{1024}r + \frac{\sigma\sqrt{\epsilon}}{128}r^2 + \frac{M_P}{6}r^3. \tag{A.26}$$

For a boundary step, $r = \sqrt{\epsilon}/(4\sigma)$ and $M_P r/6 \leq \sigma\sqrt{\epsilon}/4$. Rearranging (A.26) gives

$$P(x) - P(x+d) \geq \frac{1}{4}\sigma\sqrt{\epsilon}r^2 + \frac{\lambda}{2}r^2 - \frac{\epsilon}{1024}r,$$

which is exactly (7.14). The two endpoint value errors in (7.9) then give

$$\hat{P}(x, y) - \hat{P}(x+d, y^+) \geq \frac{61}{4096} \frac{\epsilon^{3/2}}{\sigma} + \frac{\lambda}{2}R(\sigma)^2 > \frac{\epsilon^{3/2}}{256\sigma}. \tag{A.27}$$

Thus the boundary step is accepted.

Now suppose $r < R(\sigma)$. Complementarity gives $\lambda = 0$. The common candidate-gradient estimate gives

$$\begin{aligned} \|\nabla P(x+d)\| &\leq \frac{\epsilon}{1024} + \frac{\sigma\sqrt{\epsilon}}{64}r + \frac{65}{64}\sigma\sqrt{\epsilon}r + \frac{M_P}{2}r^2 \\ &\leq \frac{1+4+260+192}{1024}\epsilon = \frac{457}{1024}\epsilon. \end{aligned}$$

Applying the positive-semidefiniteness part of (2.4) with $B = \hat{H} + (65/64)\sigma\sqrt{\epsilon}I$ and using Hessian Lipschitz continuity likewise imply

$$\begin{aligned}\lambda_{\min}(\nabla^2 P(x+d)) &\geq -\frac{65}{64}\sigma\sqrt{\epsilon} - \frac{1}{64}\sigma\sqrt{\epsilon} - M_P r \\ &\geq -\frac{162}{64}\sigma\sqrt{\epsilon} = -\frac{81}{32}\sigma\sqrt{\epsilon}.\end{aligned}$$

The validation errors in (7.11) therefore give

$$\|\hat{g}_v\| \leq 458\epsilon/1024 < \epsilon/2, \quad \lambda_{\min}(\hat{H}_v) \geq -82\sigma\sqrt{\epsilon}/32 > -(21/8)\sigma\sqrt{\epsilon}.$$

This proves (ii). Finally, if the validation test succeeds, the same error bounds give

$$\|\nabla P(x^+)\| \leq \epsilon/2 + \epsilon/1024 < \epsilon,$$

and

$$\lambda_{\min}(\nabla^2 P(x^+)) \geq -\left(\frac{21}{8} + \frac{1}{32}\right)\sigma\sqrt{\epsilon} = -\frac{85}{32}\sigma\sqrt{\epsilon}.$$

This proves (iii). □

Proof of Theorem 7.1. Set

$$\sigma_* := \sqrt{M_P/6}, \quad \bar{\sigma} := 2\sigma_* = 2\sqrt{M_P/6}. \quad (\text{A.28})$$

These quantities are positive because Assumption 5.1 assumes $\rho > 0$. Choose $\epsilon_0 \in (0, 1)$ small enough that Lemmas 7.3 and 7.4 apply and

$$\sigma_0 = \mathcal{A}_\epsilon^{-1} < \sigma_*. \quad (\text{A.29})$$

The last condition is possible because $\mathcal{A}_\epsilon \rightarrow \infty$ as $\epsilon \downarrow 0$ and the common constants give $M_P > 0$. Every restriction imposed here and in the two cited lemmas depends only on (ℓ, μ, ρ) ; hence the same ϵ_0 works for every payoff in the class and every initialization.

The scale is doubled only after a rejected boundary trial or a failed interior validation. By Lemma 7.4, once a generated scale is at least σ_* , a boundary trial is accepted and an interior trial terminates the method. The first generated scale at least σ_* is smaller than $2\sigma_*$, so no scale exceeds $\bar{\sigma}$. Counting the doublings needed to cross from $\mathcal{A}_\epsilon^{-1}$ to σ_* proves

$$J \leq \lceil \log_2(\sigma_* \mathcal{A}_\epsilon) \rceil. \quad (\text{A.30})$$

It remains to bound accepted boundary steps, including those at unsafe scales. The corrected value is not a one-sided bound on P , so we must account explicitly for changes in the stored proxy. At scale σ , (7.8) gives

$$|\hat{P} - P(x)| \leq \mathcal{E}_\epsilon(\sigma).$$

Every accepted boundary step decreases the stored proxy by at least $\epsilon^{3/2}/(256\sigma)$. The candidate tuple becomes the next stored tuple without recomputation. A failure leaves x unchanged, doubles the scale, and recomputes the current tuple. At such a refresh from σ to 2σ , the upward jump of the stored proxy is at most

$$\mathcal{E}_\epsilon(\sigma) + \mathcal{E}_\epsilon(2\sigma). \quad (\text{A.31})$$

The failure scales form a subset of $\{2^j \sigma_0 : j \geq 0\}$. Extending the sum to the whole set gives

$$\sum_{j \geq 0} \left(2^{-3j} + 2^{-3(j+1)} \right) = \frac{9}{7}. \quad (\text{A.32})$$

The initial proxy is at most $P(x_0) + \mathcal{E}_\epsilon(\sigma_0)$, and every final stored proxy is at least $P^* - \mathcal{E}_\epsilon(\sigma_0)$ because the scale never decreases. Therefore, for every finite prefix with N accepted boundary steps, enumerate those steps chronologically and let σ_k denote the scale of the k th accepted boundary trial. Then

$$\frac{\epsilon^{3/2}}{256} \sum_{k=0}^{N-1} \frac{1}{\sigma_k} \leq \Delta_P + \frac{23}{1536} c_0^3 \rho \kappa^3 \epsilon^{3/2} \mathcal{A}_\epsilon^3. \quad (\text{A.33})$$

Indeed, before substituting (7.7), the proxy-error multiplier $23/7$ is the sum of two endpoint copies and the refresh sum in (A.32). Since every $\sigma_k \leq \bar{\sigma}$, (A.33) implies

$$N \leq 256 \bar{\sigma} \Delta_P \epsilon^{-3/2} + \frac{23}{6} \bar{\sigma} c_0^3 \rho \kappa^3 \mathcal{A}_\epsilon^3. \quad (\text{A.34})$$

This uniform finite-prefix bound also proves termination. Suppose, to the contrary, that the loop is infinite. At any fixed unsafe scale, infinitely many accepted boundary steps would contradict (A.34). Hence a nonaccepted trial must eventually occur. Under the nontermination hypothesis, such a trial either fails the interior validation or rejects a boundary candidate, and therefore doubles the scale. After finitely many such doublings a safe scale is generated. At a safe scale, an interior trial terminates, a boundary rejection is impossible, and an infinite sequence of accepted boundary steps again contradicts (A.34). This contradiction proves termination. Every nonterminal trial is either an accepted boundary or a failure, and the final trial is interior, so $Q = K + J + 1$. Combining (A.30) and (A.34), now with $N = K$, gives the more explicit estimate

$$Q \leq 256 \bar{\sigma} \Delta_P \epsilon^{-3/2} + \frac{23}{6} \bar{\sigma} c_0^3 \rho \kappa^3 \mathcal{A}_\epsilon^3 + \lceil \log_2(\sigma_* \mathcal{A}_\epsilon) \rceil + 1. \quad (\text{A.35})$$

Substituting $\bar{\sigma} = 2\sqrt{M_P/6}$ proves (7.18a); the coefficient of its leading term is $512/\sqrt{6}$.

Finally, (7.16) and $\sigma \leq \bar{\sigma}$ give (7.17). Since

$$\left(\frac{85}{16\sqrt{6}} \right)^2 = \frac{7225}{1536},$$

replacing the internal tolerance by $(1536/7225)\epsilon$ gives the standard curvature threshold $-\sqrt{M_P}\epsilon$; its gradient bound is only strengthened.

It remains to count inner-gradient evaluations. Define, only for this accounting argument,

$$N_{\text{init}} := \left\lceil \log_2 \max \left\{ 1, \frac{\|\nabla_y f(x_0, y_{-1})\|}{c_0 a_0 R(\sigma_0)} \right\} \right\rceil, \quad (\text{A.36})$$

$$N_{\text{val}} := \left\lceil \log_2 \max \left\{ 1, \frac{\kappa \mathcal{A}_\epsilon}{4\epsilon} \right\} \right\rceil. \quad (\text{A.37})$$

The initial sequence of halving calls needs at most N_{init} successful calls to reach $c_0 a_0 R(\sigma_0)$. Every trust-region trial uses exactly N_w successful calls to construct its working candidate. Every failure uses one additional successful call to refresh the current tuple at the doubled scale. Hence these operations require at most

$$N_{\text{init}} + N_w Q + J$$

successful halvings.

It remains to count validation. There are at most J failed validators and one terminal validator. Immediately before validation, the candidate satisfies the working invariant, so its residual is at most

$$c_0 \ell R(\sigma) = \frac{c_0 \ell \sqrt{\epsilon}}{4\sigma}.$$

The target in (7.10) is at least $c_0\mu\epsilon^{3/2}$. Since $\sigma \geq \sigma_0 = \mathcal{A}_\epsilon^{-1}$, the ratio between the starting residual and the validation target is at most $\kappa\mathcal{A}_\epsilon/(4\epsilon)$. Therefore each validator requires at most N_{val} additional successful halvings. The total number of successful halving calls is consequently bounded by

$$N_{\text{w}}Q + J + (J+1)N_{\text{val}} + N_{\text{init}}.$$

Apply Lemma 7.1 to the entire sequence of inner objectives $q_x = -f(x, \cdot)$. They share the same (μ, ℓ) , and the initial secant obeys $\mu \leq a_0 \leq \ell$. Thus all failed curvature guesses together contribute at most $\lceil \log_4 \kappa \rceil$ attempted stages, not one such term per outer trial. Computing the initial secant and at most $J+1$ validation secants costs no more than $2(J+2)$ additional inner-gradient evaluations. Consequently,

$$N_y \leq (4 + 2\sqrt{10}C_1)\sqrt{\kappa} \left[N_{\text{w}}Q + J + (J+1)N_{\text{val}} + N_{\text{init}} + \lceil \log_4 \kappa \rceil \right] + 2(J+2). \quad (\text{A.38})$$

Finally, N_{w} is a universal integer, and

$$\begin{aligned} J &= O_{\text{fixed}}(\log \mathcal{A}_\epsilon + 1), \\ N_{\text{val}} &= O_{\text{fixed}}(\log(1/\epsilon)), \\ N_{\text{init}} &= O_{\text{fixed}}(\log(1/\epsilon)). \end{aligned}$$

The last relation uses $a_0 \geq \mu$ and $R(\sigma_0) = \sqrt{\epsilon}\mathcal{A}_\epsilon/4$. Substituting (A.35) into (A.38) proves (7.18b). Finally, $\sqrt{\kappa M_P} = \Theta(\kappa^2 \sqrt{\rho})$ gives the equivalent leading dependence stated in the theorem. $\square$

A.7 Adaptive gradient-scaled SCAR trust region

Proof of Lemma 8.1. Lipschitz continuity of the full gradient gives

$$\|\nabla_y f(x+d, y)\| \leq \|\nabla_y f(x, y)\| + \ell r \leq \ell(c_0 r_\epsilon(\sigma) + r).$$

After $N_{\text{tr}}(r, \sigma)$ successful halvings, this is at most $c_0 \ell r_\epsilon(\sigma)$ by (8.2). The case $r = 0$ requires no halving. At an unchanged x , use $r_\epsilon(2\sigma) = r_\epsilon(\sigma)/2$. $\square$

Proof of Lemma 8.2. Strong concavity and (8.1) imply

$$e := \|y - y^*(x)\| \leq \frac{c_0 \kappa \sqrt{\epsilon}}{4\sigma} \leq \frac{c_0 \kappa}{4} \sqrt{\epsilon} \mathcal{A}_\epsilon.$$

For sufficiently small ϵ , uniformly over the class, this is at most μ/ρ . The corrected-oracle estimates in Section 5 therefore give

$$\|\hat{g} - \nabla P(x)\| \leq \frac{(1+\kappa)\rho}{2} e^2, \quad |\hat{P} - P(x)| \leq \frac{7\rho}{24} e^3.$$

Substitution yields (8.4) with the constants in (8.3). The working invariant and the validation target are unchanged from Section 7, so Lemma 7.3 gives the remaining assertions. $\square$

Proof of Lemma 8.3. Apply Lemma A.1 with $F = P$, $M = M_P$,

$$a = \sigma\sqrt{h} + \frac{\sigma\sqrt{\epsilon}}{64}, \quad \xi = \frac{\sigma\sqrt{\epsilon}}{64}, \quad \delta_0 = \delta_+ = \frac{\epsilon}{1024}.$$

The endpoint errors follow from (7.9). Since $r \leq \sqrt{h}/(4\sigma)$ and $M_P \leq 6\sigma^2$,

$$\frac{M_P}{6} r^3 \leq \frac{W}{4}.$$

Thus (A.1b) gives

$$P(x) - P(x+d) \geq \frac{W}{4} + \frac{U}{2} - \frac{\epsilon r}{1024}.$$

Young's inequality and $h \geq \epsilon$ imply

$$\frac{\epsilon r}{1024} \leq \frac{W}{8} + \frac{2\epsilon^2}{1024^2 \sigma \sqrt{h}} \leq \frac{W}{8} + \frac{2T}{1024^2}.$$

After paying the two corrected-value errors,

$$\hat{P} - \hat{P}^+ \geq \frac{W}{8} + \frac{U}{2} - \left(\frac{2}{1024^2} + \frac{2}{4096} \right) T \geq \frac{W}{8} + \frac{U}{4} - \frac{T}{1024}.$$

For the gradient test, (A.1c) gives

$$\|\hat{g}^+\| \leq \lambda r + \frac{h}{4} + \frac{h}{128} + \frac{3h}{16} + \frac{h}{512} = \lambda r + \frac{229h}{512} \leq \lambda r + \frac{h}{2}.$$

Hence every ordinary trial is accepted at a safe scale.

If $h = \epsilon$ and $r < R(h, \sigma)$, the model is exactly (7.12), with an interior solution. The interior argument in the proof of Lemma 7.4 applies to the same working-accurate endpoint tuples and validation target. It gives

$$\|\hat{g}_v\| \leq \frac{458}{1024} \epsilon, \quad \lambda_{\min}(\hat{H}_v) \geq -\frac{82}{32} \sigma \sqrt{\epsilon},$$

so validation succeeds. At any generated scale, a successful validation and (7.11) imply

$$\|\nabla P(x^+)\| \leq \frac{\epsilon}{2} + \frac{\epsilon}{1024} < \epsilon, \quad \lambda_{\min}(\nabla^2 P(x^+)) \geq -\left(\frac{21}{8} + \frac{1}{32}\right) \sigma \sqrt{\epsilon}.$$

This proves (8.8). □

Lemma A.8 (Accepted-step potential decrease). *Every ordinary accepted step of Algorithm 5 satisfies*

$$\Phi - \Phi^+ \geq \frac{W}{32} + \frac{T}{1024} + \frac{U}{16}, \tag{A.39}$$

where the scale is unchanged and Φ is defined by (8.11).

Proof. First note that if $h > \epsilon$ and $h^+ \leq \max\{h/2, \epsilon\}$, then

$$\mathcal{L}_\epsilon(h^+) \leq \mathcal{L}_\epsilon(h) - 1. \tag{A.40}$$

For $\epsilon < h \leq 2\epsilon$, the two levels are 1 and 0. For $h > 2\epsilon$, the result follows from $\lceil t-1 \rceil = \lceil t \rceil - 1$. We will also use, for $c \geq 1$ and $h^+ \leq ch$,

$$\mathcal{L}_\epsilon(h^+) - \mathcal{L}_\epsilon(h) \leq \lceil \log_2 c \rceil \leq 1 + \log_2 c. \tag{A.41}$$

Suppose first that the accepted step is interior. Then $\lambda = 0$ and $U = 0$. Since this is an ordinary trial, $h > \epsilon$. The gradient acceptance test implies $h^+ \leq \max\{h/2, \epsilon\}$, so (A.40) and the value test yield

$$\Phi - \Phi^+ \geq \frac{W}{8} - \frac{T}{1024} + \frac{T}{512} = \frac{W}{8} + \frac{T}{1024}.$$

This proves (A.39) in the interior case.

For a boundary step, put $z = \lambda r/h$. The gradient test gives $h^+/h \leq \max\{1, 1/2 + z\}$. By (A.41),

$$\mathcal{L}_\epsilon(h^+) - \mathcal{L}_\epsilon(h) \leq 1 + \log_2 \max\{1, 1/2 + z\} \leq 1 + \frac{z}{\ln 2}.$$

Since $r = \sqrt{h}/(4\sigma)$,

$$W = \frac{h^{3/2}}{16\sigma} \geq \frac{T}{16}, \quad U = \frac{zh^{3/2}}{4\sigma}, \quad Tz \leq 4U.$$

Consequently,

$$\begin{aligned} \Phi - \Phi^+ &\geq \frac{W}{8} + \frac{U}{4} - \frac{T}{1024} - \frac{T}{512} \left(1 + \frac{z}{\ln 2}\right) \\ &\geq \frac{W}{8} + \left(\frac{1}{4} - \frac{1}{128 \ln 2}\right) U - \frac{3T}{1024} \\ &\geq \frac{5W}{64} + \left(\frac{1}{4} - \frac{1}{128 \ln 2}\right) U. \end{aligned}$$

Finally, $5W/64 \geq W/32 + T/1024$ and $1/4 - 1/(128 \ln 2) \geq 1/16$, proving the claim. $\square$

To account for changes of scale, define

$$\begin{aligned} E(\sigma) &:= C_v \frac{\epsilon^{3/2}}{\sigma^3}, \quad E_0 := E(\sigma_0), \quad T_0 := \frac{\epsilon^{3/2}}{\sigma_0}, \\ h_0 &:= \max\{\|\widehat{g}_0\|, \epsilon\}, \quad \mathcal{L}_0 := \mathcal{L}_\epsilon(h_0), \quad D_0 := 1 + \log_2 \left(1 + \frac{5C_g}{4\sigma_0^2}\right). \end{aligned} \tag{A.42}$$

Here $\widehat{g}_0$ is the corrected gradient after initialization. Set

$$B_\epsilon := \Delta_P + \frac{23}{7} E_0 + \frac{T_0}{512} (\mathcal{L}_0 + D_0). \tag{A.43}$$

Choose ϵ small enough that the working-error and safe-scale lemmas apply and $\sigma_0 < \sigma_* := \sqrt{M_P/6}$. Every failed trial doubles σ , and [Lemma 8.3](#) excludes failures at or above σ_* . Thus every finite prefix satisfies

$$J \leq \lceil \log_2(\sigma_* \mathcal{A}_\epsilon) \rceil, \quad \sigma \leq \bar{\sigma} := 2\sigma_*. \tag{A.44}$$

Lemma A.9 (Cumulative accepted-step budget). *Under the preceding conditions, every finite prefix satisfies*

$$\sum_{\text{accepted}} \left(\frac{W_k}{32} + \frac{T_k}{1024} + \frac{U_k}{16} \right) \leq B_\epsilon. \tag{A.45}$$

If K denotes its number of accepted steps, then

$$K \leq 1024 \bar{\sigma} B_\epsilon \epsilon^{-3/2}, \quad \sum_{\text{accepted}} \sigma_k^2 r_k^3 \leq 8B_\epsilon. \tag{A.46}$$

Every stored iterate satisfies $P(x) - P^* \leq B_\epsilon$, and

$$B_\epsilon = \Delta_P + O_{\text{fixed}}(\epsilon^{3/2} \mathcal{A}_\epsilon^3). \tag{A.47}$$

Proof. At a refresh, x is unchanged and the scale goes from σ to 2σ . By [\(8.4\)](#), the corrected-value jump is at most

$$E(\sigma) + E(2\sigma) = \frac{9}{8} E(\sigma).$$

The scales at failures form a doubling sequence, so the sum of all positive value jumps is at most

$$\frac{9}{8} E_0 \sum_{j \geq 0} 2^{-3j} = \frac{9}{7} E_0.$$

For the gradient-level term, the two corrected-gradient errors give

$$\|\widehat{g}_{\text{new}} - \widehat{g}_{\text{old}}\| \leq \frac{5C_g \epsilon}{4\sigma^2}.$$

The map $g \mapsto \max\{\|g\|, \epsilon\}$ is 1-Lipschitz and $h_{\text{old}} \geq \epsilon$. Hence

$$\frac{h_{\text{new}}}{h_{\text{old}}} \leq 1 + \frac{5C_g}{4\sigma^2}, \quad \mathcal{L}_\epsilon(h_{\text{new}}) - \mathcal{L}_\epsilon(h_{\text{old}}) \leq D_0.$$

Write $\mathcal{L}_{\text{old}}$ and $\mathcal{L}_{\text{new}}$ for these two levels. Since $T_{\text{new}} = T_{\text{old}}/2$ and the levels are nonnegative, the positive jump of the level contribution is at most

$$\left[\frac{T_{\text{old}}}{1024} (\mathcal{L}_{\text{new}} - 2\mathcal{L}_{\text{old}}) \right]_+ \leq \frac{T_{\text{old}} D_0}{1024}.$$

Its cumulative positive jump is therefore bounded by

$$\frac{T_0 D_0}{1024} \sum_{j \geq 0} 2^{-j} = \frac{T_0 D_0}{512}.$$

The initial potential is at most $P(x_0) + E_0 + T_0 \mathcal{L}_0/512$, and every stored potential is at least $P^* - E_0$. Telescoping (A.39) and paying the two refresh bounds gives

$$\sum_{\text{accepted}} \left(\frac{W}{32} + \frac{T}{1024} + \frac{U}{16} \right) \leq \Delta_P + 2E_0 + \frac{9}{7}E_0 + \frac{T_0}{512}(\mathcal{L}_0 + D_0) = B_\epsilon.$$

The same upper bound on the stored potential, together with $P(x) \leq \Phi + E_0$, gives $P(x) - P^* \leq B_\epsilon$.

By (A.44), $T_k \geq \epsilon^{3/2}/\bar{\sigma}$, which proves the bound on K . The radius constraint also gives

$$4\sigma_k^2 r_k^3 \leq \sigma_k \sqrt{h_k} r_k^2 = W_k.$$

Combining this with (A.45) proves the cubic-path bound.

Finally,

$$E_0 = C_v \epsilon^{3/2} \mathcal{A}_\epsilon^3, \quad T_0 = \epsilon^{3/2} \mathcal{A}_\epsilon.$$

At the initial stored point,

$$\|\widehat{g}_0\| \leq \|\nabla P(x_0)\| + C_g \epsilon \mathcal{A}_\epsilon^2.$$

For each fixed payoff and initialization this implies $\mathcal{L}_0 = O_{\text{fixed}}(\mathcal{A}_\epsilon)$, while $D_0 = O_{\text{fixed}}(1 + \log \mathcal{A}_\epsilon)$. Substitution into (A.43) yields (A.47). $\square$

Proof of Theorem 8.1. Choose ϵ_0 so that Lemmas 8.2 and 8.3 hold and $\mathcal{A}_\epsilon^{-1} < \sqrt{M_P/6}$ for $0 < \epsilon \leq \epsilon_0$. These restrictions depend only on the common constants (ℓ, μ, ρ) .

Each inner sequence is finite by Lemma 7.1 and the positive residual targets. Every nonterminal outer trial is either accepted or increases the scale. The finite-prefix bounds (??) bound both possibilities, so the algorithm terminates. With K accepted steps and J failures, $Q = K + J + 1$, and hence

$$Q \leq 1024\bar{\sigma} B_\epsilon \epsilon^{-3/2} + \lceil \log_2(\sigma_* \mathcal{A}_\epsilon) \rceil + 1.$$

Using (A.47) and $\bar{\sigma} = 2\sqrt{M_P/6}$ gives (8.10a). The output bound (8.9) follows from (8.8) and the scale cap. The tolerance conversion follows from $(85/(16\sqrt{6}))^2 = 7225/1536$.

It remains to count successful halvings. On an accepted step, set $z_k = \sigma_k r_k / \sqrt{\epsilon}$. Since $c_0 = 2^{-12}$,

$$N_{\text{tr},k} = \lceil \log_2(1 + 2^{14} z_k) \rceil \leq 16 + z_k^3.$$

For $0 \leq z \leq 1$, the ceiling is at most 15. For $z \geq 1$, use $1 + 2^{14} z \leq 2^{15} z$ and $\log_2 z \leq z^3$. Thus (A.46) gives

$$\begin{aligned} \sum_{\text{accepted}} N_{\text{tr},k} &\leq 16K + \epsilon^{-3/2} \sum_{\text{accepted}} \sigma_k^3 r_k^3 \\ &\leq 16K + 8\bar{\sigma} B_\epsilon \epsilon^{-3/2} \\ &\leq 16392\bar{\sigma} B_\epsilon \epsilon^{-3/2}. \end{aligned} \tag{A.48}$$

To bound tracking at failed trials, put $L_P = (1 + \kappa)\ell$. Smoothness and the lower bound P^* imply

$$\|\nabla P(x)\|^2 \leq 2L_P(P(x) - P^*).$$

At every stored iterate, [Lemma A.9](#) and [\(8.4\)](#) therefore give

$$h \leq \bar{h}_\epsilon := \max \left\{ \epsilon, \sqrt{2L_P B_\epsilon} + C_g \epsilon \mathcal{A}_\epsilon^2 \right\} = O_{\text{fixed}}(1).$$

Since $r \leq R(h, \sigma)$, each failed trial requires at most

$$N_{\text{fail}} := \left\lceil \log_2 \left(1 + \frac{1}{c_0} \sqrt{\frac{\bar{h}_\epsilon}{\epsilon}} \right) \right\rceil = O_{\text{fixed}}(\mathcal{A}_\epsilon)$$

working halvings. A terminal trial has $h = \epsilon$ and $r < r_\epsilon(\sigma)$, so its working tracking uses at most $\lceil \log_2(1 + 1/c_0) \rceil = 13$ halvings. Each failure uses one additional halving for the refresh.

Immediately before a validation, the residual is at most $c_0 \ell \sqrt{\epsilon}/(4\sigma)$. The validation target is at least $c_0 \mu \epsilon^{3/2}$, so the number of additional halvings is at most

$$N_{\text{val}} := \left\lceil \log_2 \max \left\{ 1, \frac{\kappa \mathcal{A}_\epsilon}{4\epsilon} \right\} \right\rceil = O_{\text{fixed}}(\mathcal{A}_\epsilon).$$

There are at most $J + 1$ validations. Initialization uses at most

$$N_{\text{init}} := \left\lceil \log_2 \max \left\{ 1, \frac{\|\nabla_y f(x_0, y_{-1})\|}{c_0 a_0 r_\epsilon(\sigma_0)} \right\} \right\rceil = O_{\text{fixed}}(\mathcal{A}_\epsilon)$$

halvings. Consequently the total number of successful halvings obeys

$$N_{\text{half}} \leq \sum_{\text{accepted}} N_{\text{tr},k} + J N_{\text{fail}} + J + 13 + (J + 1) N_{\text{val}} + N_{\text{init}}. \quad (\text{A.49})$$

By [\(A.44\)](#), $J = O_{\text{fixed}}(1 + \log \mathcal{A}_\epsilon)$. Equations [\(??\)](#) therefore imply

$$N_{\text{half}} = O(\Delta_P \sqrt{M_P} \epsilon^{-3/2}) + O_{\text{fixed}}(\mathcal{A}_\epsilon^3).$$

All inner objectives $q_x = -f(x, \cdot)$ share the same (μ, ℓ) . Apply [Lemma 7.1](#) once to the complete sequence, including rejected candidates, refreshes, and validators. The initial and validation secants require at most $2(J + 2)$ additional gradient evaluations, giving

$$N_y \leq (4 + 2\sqrt{10} C_1) \sqrt{\kappa} (N_{\text{half}} + \lceil \log_4 \kappa \rceil) + 2(J + 2). \quad (\text{A.50})$$

This proves [\(8.10b\)](#). Finally, $\sqrt{M_P} = \Theta(\kappa^{3/2} \sqrt{\rho})$ and $\sqrt{\kappa M_P} = \Theta(\kappa^2 \sqrt{\rho})$ give the stated leading dependence. $\square$

B Scope of the parameter-free guarantees

This section distinguishes the class-uniform asymptotic guarantees from a residual-only certificate that would be uniform without a positive common strong-concavity lower bound. It also clarifies the additional book-keeping used by the fixed-radius construction.

B.1 The residual-only obstruction

A parameter-free inner method such as SCAR can return a point satisfying the prescribed and directly verifiable condition $\|\nabla_y f(x, y)\| \leq \tau$. This fact alone does not provide a certificate for the value or gradient of $P(x) = \max_y f(x, y)$ that is uniform over problem classes having no positive common lower bound on the strong-concavity modulus. The issue is not merely a loss in the complexity bound: an outer stopping test based only on a residual-corrected oracle can be false.

The following scalar family shows that this difficulty persists even for the corrected value (5.11). Let

$$c := \frac{1}{2}, \quad \psi(t) := \frac{t^2}{2} + c(1 - \cos t), \quad A := \frac{1 - c}{2\pi^2},$$

and, for $0 < \mu \leq 1/4$, define

$$f_\mu(x, y) := A \tanh(x/A) (1 - \cos(2\pi\sqrt{\mu}y)) - \psi(2\sqrt{\mu}y - 1). \quad (\text{B.1})$$

Because $1 - c \leq \psi''(t) \leq 1 + c$,

$$\partial_{yy}^2 f_\mu(x, y) = 4A\pi^2\mu \tanh(x/A) \cos(2\pi\sqrt{\mu}y) - 4\mu\psi''(2\sqrt{\mu}y - 1) \leq -2(1 - c)\mu = -\mu.$$

Thus $f_\mu(x, \cdot)$ is globally μ -strongly concave. Its second derivatives are uniformly bounded for $0 < \mu \leq 1/4$. Its third derivatives are also uniformly bounded: the four distinct block types have bounds $12/A^2$, $2\pi/A$, π^2 , and $A\pi^3 + c$, respectively. Hence the joint gradient-smoothness and Hessian-Lipschitz constants admit finite upper bounds independent of μ . Moreover,

$$P_\mu(x) \geq f_\mu\left(x, \frac{1}{2\sqrt{\mu}}\right) = 2A \tanh(x/A) \geq -2A,$$

so the value function is bounded below. At $x = 0$ and $y = 1/(2\sqrt{\mu})$, $-\partial_{yy}^2 f_\mu(x, y) = 4\mu\psi''(0) = 6\mu$. Consequently, the largest valid global strong-concavity modulus lies between μ and 6μ and genuinely tends to zero along the family.

At $x = 0$, the exact maximizer is $y_\mu^*(0) = 1/(2\sqrt{\mu})$. At the point $y = 0$, however,

$$|\partial_y f_\mu(0, 0)| = 2\sqrt{\mu} |\psi'(-1)| = (2 + \sin 1)\sqrt{\mu} \longrightarrow 0, \quad \hat{g}_\mu(0, 0) = 0, \quad \hat{H}_\mu(0, 0) = 0.$$

In contrast, Danskin's theorem gives

$$P'_\mu(0) = \partial_x f_\mu\left(0, \frac{1}{2\sqrt{\mu}}\right) = 1 - \cos \pi = 2.$$

The corrected value is wrong by a fixed amount as well. Namely, $P_\mu(0) = 0$, whereas

$$\hat{P}_\mu(0, 0) = -\psi(-1) + \frac{\psi'(-1)^2}{2\psi''(-1)} \approx 0.0647, \quad (\text{B.2})$$

independently of μ . Therefore a residual tending to zero does not uniformly control either the corrected value or the corrected gradient over the union of these parameter classes. The smoothness constants remain bounded while the best strong-concavity modulus tends to zero.

Consequently, choosing SCAR as a residual module removes the inner solver's dependence on ℓ and μ , but the residual alone does not yield a single certificate valid across all positive strong-concavity moduli. Nor does a prescribed finite sequence of positive tolerances repair the issue. For any finite list $\tau_0, \dots, \tau_J > 0$, choose

$$0 < \mu < \min \left\{ \frac{1}{4}, \left(\frac{\min_j \tau_j}{2 + \sin 1} \right)^2 \right\}$$

and repeat each inner call at the outer point $x = 0$, initialized at $y = 0$. Every residual test is then already satisfied and may return the same false outer derivatives. This observation covers any predetermined finite number of tolerance-halving phases. It does not claim that indefinite refinement cannot eventually move the inner iterate; it shows that a finite prescribed refinement depth is not a certificate.

There is no contradiction with the class-uniform result in [Theorem 6.2](#). Once a positive common value of μ is fixed, the condition

$$\epsilon^2 < (2 + \sin 1)\sqrt{\mu}$$

excludes $y = 0$ as a legal return of an inner call with residual tolerance ϵ^2 . More generally, every legal return satisfies

$$\|y - y_\mu^*(x)\| \leq \frac{\epsilon^2}{\mu}.$$

Thus one threshold may be chosen uniformly after the common constants (ℓ, μ, ρ) have been fixed. What fails is a threshold uniform over their union as the common lower bound μ tends to zero.

SCAR also does not return a certified lower bound on μ , a certified inner optimality gap, or an interval containing $P(x)$. Its returned smoothness quantity is a line-search estimate satisfying the tested local inequalities, not a certified global upper bound on ℓ . Likewise, a successful residual reduction does not certify that the current strong-convexity guess is below the true μ . A fully observable finite-accuracy adaptive minimax certificate therefore needs additional information, such as a certified lower bound on μ , a certified inner-gap bound, genuine intervals containing P , ∇P , and $\nabla^2 P$, compactness plus a dual-gap certificate, or additional problem structure. [Theorem 6.2](#) instead gives an unknown class-uniform threshold $\epsilon_0(\ell, \mu, \rho)$; it makes neither a claim uniform as $\mu \downarrow 0$ nor an a posteriori claim that the algorithm can recognize entry into this regime.

B.2 Additional qualifications for the fixed-radius method

The class-uniform quantifier in [Theorem 7.1](#) has the same nonobservability caveat: the algorithm uses no unknown problem parameter, but it cannot observe whether $\epsilon \leq \epsilon_0(\ell, \mu, \rho)$. The proof uses the positive common lower bound on μ to convert working and validation residuals into distances and corrected-oracle errors.

The counterexample in [Section B.1](#) concerns prescribed absolute residual tolerances. It does not, by itself, rule out a threshold uniform over $\mu \downarrow 0$ for the present fixed-radius construction, whose working and validation targets are scaled by observable secants. Accordingly, [Theorem 7.1](#) asserts uniformity only after (ℓ, μ, ρ) have been fixed and makes no claim either way about a stronger union-uniform theorem.

The fixed-radius construction handles the function-value difficulty in a specific, nonstandard way. It never asserts that $\hat{P}$ is a lower or an upper bound on P . Unsafe accepted boundary steps need not decrease the true value function. Instead, the proof telescopes the stored corrected proxy and explicitly pays for every proxy jump caused by a scale refresh; see [\(A.31\)–\(A.33\)](#). This argument relies on three implementation details: the scale is nondecreasing, an accepted candidate inner point and corrected value are retained without further inner refinement, and a rejected candidate is discarded before the current tuple is refreshed. Without these bookkeeping rules, the displayed telescoping argument no longer applies without additional error accounting.

The high-accuracy validator is invoked only after an interior fixed-radius solution. Its auxiliary residual $c_0 a_x \epsilon^{3/2}$ controls inner accuracy only; it is not used as the trust-region scale or as the Hessian stopping threshold. Under the theorem’s class-uniform small- ϵ hypotheses, the validator rules out false termination, but it does not reveal whether those hypotheses hold. A doubling trick in ϵ does not by itself identify the unknown onset of this asymptotic regime. The present analysis therefore does not furnish an observable all- ϵ certificate. Establishing one would require a new certificate or argument, for example a certified lower

bound on μ , a certified inner-gap or value-function interval, compactness with a computable dual gap, or other problem-specific structure.

References

- [1] Yuntian Jiang, Chang He, Chuwen Zhang, Dongdong Ge, Bo Jiang, and Yinyu Ye. Beyond nonconvexity: A universal trust-region method with new analyses. *Journal of Scientific Computing*, 2026. arXiv:2311.11489.
- [2] Fadi Hamad and Oliver Hinder. A consistently adaptive trust-region method. *Advances in Neural Information Processing Systems*, 35:6640–6653, 2022.
- [3] Shaoying Fang, Naoki Marumo, and Akiko Takeda. Parameter-free cubic-regularized newton method: Sharp complexity and generalized smoothness. *arXiv preprint arXiv:2607.10741*, 2026.
- [4] Luo Luo, Yujun Li, and Cheng Chen. Finding second-order stationary points in nonconvex-strongly-concave minimax optimization. In *Advances in Neural Information Processing Systems*, volume 35, pages 36667–36679, 2022.
- [5] Junlin Wang and Zi Xu. Gradient norm regularization second-order algorithms for solving nonconvex-strongly concave minimax problems. *Journal of Scientific Computing*, 105(2):41, 2025. doi: 10.1007/s10915-025-03069-8.
- [6] Yurii Nesterov. *Introductory Lectures on Convex Optimization: A Basic Course*, volume 87 of *Applied Optimization*. Kluwer Academic Publishers, Boston, 2004. doi: 10.1007/978-1-4419-8853-9.
- [7] David G. Luenberger and Yinyu Ye. *Linear and Nonlinear Programming*. Springer, Cham, 5 edition, 2021. doi: 10.1007/978-3-030-85450-8.
- [8] Coralia Cartis, Nicholas I. M. Gould, and Philippe L. Toint. Adaptive cubic regularisation methods for unconstrained optimization. part ii: Worst-case function- and derivative-evaluation complexity. *Mathematical Programming*, 130(2):295–319, 2011.
- [9] Guanghui Lan, Yuyuan Ouyang, and Zhe Zhang. Optimal and parameter-free gradient minimization methods for convex and nonconvex optimization. *Mathematical Programming*, 2026. doi: 10.1007/s10107-026-02352-2.
- [10] Yuanhao Wang and Jian Li. Improved algorithms for convex-concave minimax optimization. In *Advances in Neural Information Processing Systems*, volume 33, pages 4800–4810, 2020.